

SECTION A

THE COMPLETE BOOK

General Awareness - Reasoning - Mathematics
English - Hindi

The 100 marks that every DSSSB post shares - TGT, PGT, PRT, LDC, DASS, JE, ASO and more

5 sections x 20 questions = 100 marks

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WHY SECTION A DECIDES YOUR SELECTION

Section A is 100 of the 200 marks - exactly half the paper.

It is the SAME for every post, so one book serves TGT, PGT, PRT, LDC and the rest.

It is far more predictable than the subject paper: the same topics repeat every year.

Most candidates over-prepare their subject and lose 25-30 marks here. That is the gap.

Pattern verified against official DSSSB question booklets published on dsssb.delhi.gov.in. Sources are cited in Chapter 1.1.

Current affairs must be updated from a daily source - see the honest note in Chapter 3.1 before you rely on that Part.

Hindi is delivered in Devanagari as a separate printable file - see Chapter 7.1 for why and where.

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PART 1

THE SECTION A PATTERN AND YOUR STRATEGY

100 marks, five sections, twenty questions each. Identical for every DSSSB post. Read this Part before you study a single topic.

1.1 The Pattern, Verified

1.1.1 What Section A actually is

Every DSSSB recruitment paper - TGT, PGT, PRT, LDC, DASS, Junior Engineer, ASO, Nursing Officer, Scientific Assistant, Special Educator - shares one common half. That half is **Section A**.

Section A component	Questions	Marks
General Awareness	20	20
General Intelligence and Reasoning Ability	20	20
Arithmetical and Numerical Ability	20	20
Test of English Language and Comprehension	20	20
Test of Hindi Language and Comprehension	20	20
TOTAL SECTION A	100	100

Whole paper	Marks
Section A (the five components above)	100
Section B (post-specific subject + teaching methodology, or the technical paper)	100
TOTAL	200

Duration 2 hours (120 minutes) for 200 questions. That is **36 seconds per question** on average.

Marking +1 for a correct answer, -0.25 for a wrong answer, 0 for a skipped question.

Mode Computer Based Test (CBT), objective multiple choice with four options.

Language Bilingual (English and Hindi), except the two language-test sections themselves.

How this pattern was verified

This is not taken from a coaching blog. I checked it against **official DSSSB question booklets hosted on dsssb.delhi.gov.in**, which print the section table on the cover page. One such booklet lists "General Awareness 20", "General Intelligence and Reasoning Ability 20" and so on, against "Maximum Marks: 200" and "Time Allowed: 2 hours". Another official paper file shows the per-question marking directly: **Correct Marks: 1, Wrong Marks: 0.25**.

Primary sources (official): [DSSSB official question booklet - dsssb.delhi.gov.in](https://dsssb.delhi.gov.in/sites/default/files/DSSSB/universal-tab/qb12014.pdf) , [DSSSB official paper showing per-question marking](https://dsssb.delhi.gov.in/sites/default/files/DSSSB/universal-tab/25thnov21_s1.pdf) , [DSSSB Scheme and Syllabus page](https://dsssb.delhi.gov.in/dsssb/scheme-syllabus).

Secondary sources consulted for the topic breakdown: [DSSSB TGT syllabus - pw.live](https://www.pw.live/teaching/exams/dsssb-tgt-syllabus) , [DSSSB PRT syllabus - pw.live](https://www.pw.live/teaching/exams/dsssb-prt-syllabus) , [DSSSB syllabus overview - pw.live](https://www.pw.live/ssc/exams/dsssb-syllabus) , [DSSSB Statistical Assistant pattern - testbook](https://testbook.com/dsssb-statistical-assistant/syllabus-exam-pattern) , [DSSSB Scientific Assistant pattern - testbook](https://testbook.com/dsssb-scientific-assistant/syllabus-exam-pattern) , [DSSSB JE syllabus topic list - byjus](https://byjus.com/govt-exams/dsssb-je-syllabus/) , [DSSSB syllabus - byjus](https://byjus.com/govt-exams/dsssb-syllabus/). Content from these sources was paraphrased and condensed for compliance with licensing restrictions.

Still verify the notification for your own cycle. Some posts are two-tier, a few have a qualifying Tier 1 with merit decided by Tier 2, and sectional qualifying marks sometimes apply to Section B. The 20+20+20+20+20 structure of Section A has been remarkably stable, but the wrapper around it varies by post.

1.1.2 Why Section A is where selections are won

Think about what these two halves are.

Section B is your subject. You studied it for three years at degree level. Every other candidate in the hall also studied it for three years. The marks there cluster tightly - most serious candidates score somewhere between 60 and 80, and pushing from 70 to 75 costs enormous effort.

Section A is the opposite. It is broad, shallow and highly repetitive. And most candidates barely touch it, because it feels less serious than their subject. The spread here is enormous - I have seen candidates score 35 and candidates score 88 on the same paper.

That spread is the selection. A candidate who scores 70 in Section B and 85 in Section A beats a candidate who scores 80 in Section B and 55 in Section A - by 20 marks.

Reality	Consequence for you
Section A is exactly half the paper	It deserves close to half your preparation time. Almost nobody gives it that
It is identical across posts	Study it once and it serves any DSSSB exam you ever sit
Four of the five sections are pure technique or pure recall	They are trainable in weeks, unlike a subject which takes years
The topics repeat with unusual fidelity	Last year's topic list is next year's topic list, with the current affairs swapped out

1.1.3 What "95+" actually requires

You asked for a book that gets a candidate past 95. Let me be precise and honest about what that means, because a vague promise would not help you.

95 out of 100 in Section A is possible but demanding. Here is the realistic route:

Section	Target	Why this target is realistic
Reasoning	19-20 / 20	Pure technique. With practice, near-full marks is normal. There is no "unknown" in reasoning - every question type is finite and learnable
Mathematics	18-20 / 20	Also technique. The topic list is closed and the difficulty is Class 10 level. Speed is the only real constraint
English	17-19 / 20	Grammar and error-spotting are rule-based. Vocabulary is the one open-ended part
Hindi	17-19 / 20	Same as English - vyakaran is rule-based and scoring
General Awareness	13-17 / 20	The honest weak point. Static GK is learnable, but current affairs is open-ended. Nobody reliably scores 20 here

Add the realistic middles: $19 + 19 + 18 + 18 + 15 = 89$. Add the optimistic ends: $20 + 20 + 19 + 19 + 17 = 95$.

The honest version of the "95+" promise

Reasoning, Maths, English and Hindi are 80 marks, and they are almost entirely within your control. They are rule-based and finite. If you work through Parts 4, 5, 6 and 7 of this book properly and practise, scoring **74-78 out of those 80** is a realistic, repeatable outcome. That is the real engine of a high score.

General Awareness is 20 marks and only partly within your control. Static GK - history, geography, polity, economy, science - is learnable and this book covers it thoroughly. **Current affairs is not learnable from any book**, including this one, because the events have not happened yet when the book is written. You must add a daily source. I explain exactly how in Chapter 3.1, and I will not pretend otherwise.

So the honest promise is this: **this book can take you to about 90, and a disciplined current-affairs habit on top of it can take you past 95.** Any book that claims to do the whole 95 by itself, current affairs included, is not being straight with you.

1.2 Time Strategy Inside the Exam Hall

1.2.1 The 36-second problem

200 questions in 120 minutes is 36 seconds each. But the questions are not equal. A synonym takes 8 seconds. A time-and-work sum takes 90. A reading comprehension passage takes 4 minutes for 5 questions.

So you do not budget per question. You budget **per section**.

Section	Questions	Suggested time	Reason
Hindi Language	20	10 minutes	Fastest for most Indian candidates. Bank these marks first
English Language	20	12 minutes	Grammar and vocabulary are instant; only the passage takes time
General Awareness	20	8 minutes	You either know it or you do not. Never think about a GA question - answer or skip in 20 seconds
Reasoning	20	20 minutes	Some types (puzzles, series) genuinely need working time
Mathematics	20	25 minutes	The slowest section. Do the formula-based ones first, leave the word problems
Section A subtotal	100	75 minutes	-
Section B (your subject)	100	42 minutes	Your subject is recall-heavy, so it moves faster than maths
Buffer / review	-	3 minutes	Check for OMR or marking errors

The order to attempt in

Do **not** attempt the paper top to bottom. Attempt it in **increasing order of time cost**:

Hindi, then English, then General Awareness, then your Section B subject, then Reasoning, then Mathematics.

Two reasons. First, you bank the certain marks before fatigue and clock-panic set in. Second, if you run out of time - which most candidates do - the questions you sacrifice are the slow ones you were least likely to get right anyway, not the fast ones you were certain about.

1.2.2 The two-pass method

How to move through a section

PASS 1 (fast sweep)

- Read each question once.
- If the answer arrives within your time budget, mark it and move on.
- If it does not, FLAG IT and move on immediately. Do not linger.
- Target: finish pass 1 using about 70% of the section's time.

PASS 2 (return sweep)

- Go back only to the flagged questions.
- For each, try to ELIMINATE two options.
- Succeeded? Answer it - the expected value is positive.
- Failed? Leave it blank and move on.

NEVER let one question consume more than 2 minutes. Ever.

1.2.3 The mathematics of guessing - learn this table

With +1 and -0.25, guessing is not free but it is not forbidden either. The decision is arithmetic, not courage.

Your situation	Chance of being right	Expected marks	Decision
You know it	100%	+1.00	Answer
You eliminated 3 of 4 (certain of two, unsure which)	50%	$0.5(1) + 0.5(-0.25) = +0.375$	Answer - clearly worth it
You eliminated 2 of 4	33%	$0.33(1) + 0.67(-0.25) = +0.16$	Answer - still positive
You eliminated nothing (blind guess)	25%	$0.25(1) + 0.75(-0.25) = +0.0625$	Marginal. Skip if any distractor looks deliberately attractive
You have a hunch but cannot eliminate anything	below 25%	Negative	Skip

The rule that follows: never leave blank a question where you can eliminate two options. Never attempt a question where you can eliminate none.

Why blind guessing is worse than the maths suggests

The +0.0625 figure assumes each of the four options is equally likely - a random lottery. Real exam papers are not random. Examiners deliberately build **attractive wrong options** that match a common misconception. So on a question you genuinely do not understand, your real chance of being right is often *below* 25%, because you will be drawn to the trap rather than picking uniformly.

That is why the practical rule is: **eliminate, or skip**. And it is why the "why the wrong options are wrong" explanations throughout this book matter more than the answers themselves - they are training your elimination instinct.

1.3 How to Use This Book

1.3.1 The study sequence

Week	Focus	Target
1-2	Part 5 (Reasoning) - all chapters, every type	Recognise every question type on sight
3-4	Part 4 (Mathematics) - formulas first, then chapters	Learn every formula in Chapter 4.1 by heart
5	Part 6 (English) - grammar rules, then vocabulary	Finish all rule chapters
6	Part 7 (Hindi) - vyakaran	Finish all rule chapters
7-9	Part 2 (General Awareness) - one chapter per two days	One full pass of static GK
10	Part 3 (Current Affairs) - set up your daily habit	Habit running, notes started
11-12	Revision: Part 8 only, plus daily current affairs	Two full passes of Part 8
Ongoing from week 1	15 minutes of current affairs every single day	Non-negotiable

1.3.2 The rules that make this work

- Reasoning and Maths need a pen, not just eyes.** Reading a solved example teaches you nothing. Cover the solution, attempt it, then compare. This is the single biggest difference between candidates who score 19 and candidates who score 12.
- Keep a leak list.** One line per mistake, naming the **concept**, not the question. "Forgot that percentage change uses the original as base." After each Part, that list is your revision syllabus.
- Static GK needs spaced repetition, not long sessions.** Twenty minutes daily beats four hours weekly. Facts decay; revisit them.
- Start current affairs on day one, not month three.** Six months of 15-minute daily reading is worth more than two weeks of panic revision, because current affairs questions span the whole year.
- Time yourself from week 3.** Accuracy without speed fails a 36-second-per-question paper. Practise chapters against a clock.

1.3.3 What the labels in this book mean

Label	Meaning
[PYQ-TYPE]	A question of a type that demonstrably recurs in DSSSB Section A papers. The wording is mine; the pattern is taken from the official papers and syllabus
[EXPECTED]	A practice question I wrote on a syllabus topic, in the same style and difficulty
[FORMULA]	A formula or rule to commit to memory. These are the load-bearing facts
[SHORTCUT]	A faster method than the textbook one. Learn these - they buy you the time you need
[TRAP]	A commonly mis-answered item. Read it twice

An honest statement about sources, before you begin

I have verified the **structure** of Section A against official DSSSB question booklets, and the **topic lists** against the published syllabus and multiple secondary sources, all cited in Chapter 1.1.

What I do **not** have is a complete archive of every DSSSB question ever asked across TGT, PGT, PRT, LDC and the rest. The official site hosts scanned booklets, but they are not machine-readable in bulk from here.

So I have not labelled anything "asked in DSSSB PRT 2019" unless I could support it. Instead, questions are marked [PYQ-TYPE] where the *pattern* is established, and [EXPECTED] where I built the question myself. The concepts and question **types** are what repeat across these papers - and those I have covered exhaustively. Misattributing a question would make this book look more authoritative while making it less trustworthy, and that trade is not worth making for something you are staking a career on.

PART 2

GENERAL AWARENESS - STATIC GK

20 marks, shared with Current Affairs. Static GK is the learnable half - history, geography, polity, economy, science and Delhi-specific questions. This Part is the largest in the book because the syllabus genuinely is that wide.

2.1 How to Study This Part

2.1.1 What the syllabus says, and what that means

The official DSSSB syllabus describes General Awareness as testing awareness of the environment around the candidate and its application to society, including **current events** and matters of **everyday observation and experience** in their scientific aspect, with questions from **History, Geography, General Science, Economics, Polity and scientific research**.

That is deliberately broad. But real papers concentrate heavily, and the concentration is predictable.

Area	Typical questions per paper	Priority
Current Affairs (see Part 3)	5-8	CRITICAL
Indian Polity and Constitution	2-3	HIGH
Indian History	2-3	HIGH
Geography (India + World)	2-3	HIGH
General Science	2-3	HIGH
Indian Economy	1-2	MEDIUM
Static miscellany (awards, sports, days, organisations)	1-2	MEDIUM
Delhi-specific GK	0-2	MEDIUM (DSSSB is a Delhi body)

How to actually retain this Part

Static GK is defeated by **spaced repetition**, not by long sessions. Twenty minutes daily beats four hours on a Sunday, because facts decay on a predictable curve and each revisit flattens that curve.

Practical method: read one chapter, then re-read only its **tables** on day 3, day 10 and day 30. The tables in this Part were written to be revisable in isolation for exactly this reason.

2.2 Indian Polity and Constitution

2.2.1 The Constitution - core facts

Fact	Detail
Constituent Assembly formed	1946, under the Cabinet Mission Plan
Chairman of the Constituent Assembly	Dr. Rajendra Prasad
Chairman of the Drafting Committee	Dr. B. R. Ambedkar - called the Father of the Indian Constitution
Constitution adopted	26 November 1949 (celebrated as Constitution Day / Samvidhan Divas)
Constitution came into force	26 January 1950 (Republic Day)
Time taken	2 years, 11 months and 18 days
Original Constitution	395 Articles, 22 Parts, 8 Schedules
Presently	Around 470 Articles, 25 Parts, 12 Schedules
Longest written constitution in the world	Yes
Borrowed features	Parliamentary system and rule of law from Britain ; Fundamental Rights, judicial review and impeachment from the USA ; Directive Principles from Ireland ; Fundamental Duties and five-year plans from the USSR ; Emergency provisions from Germany ; concurrent list from Australia ; federation with a strong centre from Canada

2.2.2 Fundamental Rights, Duties and Directive Principles

Right	Articles
Right to Equality	14 to 18
Right to Freedom	19 to 22
Right against Exploitation	23 to 24
Right to Freedom of Religion	25 to 28
Cultural and Educational Rights	29 to 30
Right to Constitutional Remedies	32 - called the "heart and soul of the Constitution" by Ambedkar

Originally seven Fundamental Rights The **Right to Property** was removed by the **44th Amendment (1978)** and made an ordinary legal right under Article 300A. Six remain.

Article 21A Right to Education for children aged 6-14, added by the **86th Amendment (2002)**.

Fundamental Duties Added by the **42nd Amendment (1976)** on the recommendation of the **Swaran Singh Committee**. Originally 10; an 11th (duty of a parent to provide education) was added by the **86th Amendment**. Listed in **Article 51A**, Part IVA.

Directive Principles Part IV, Articles 36-51. Borrowed from **Ireland**. They are **not enforceable** in a court.

The five writs **Habeas Corpus** (produce the body), **Mandamus** (we command), **Prohibition**, **Certiorari**, **Quo Warranto**.

2.2.3 Important Articles worth memorising

Article	Subject
1	India, that is Bharat, shall be a Union of States
14	Equality before law
19	Six freedoms including speech and expression
21	Protection of life and personal liberty
21A	Right to Education
32	Right to constitutional remedies (Supreme Court)
40	Organisation of village panchayats
44	Uniform Civil Code
51A	Fundamental Duties
72	President's power to grant pardons
110	Definition of a Money Bill
123	President's ordinance power
143	President's power to consult the Supreme Court
226	High Court's writ jurisdiction
280	Finance Commission
300A	Right to Property
352	National Emergency
356	President's Rule in a state
360	Financial Emergency
368	Amendment of the Constitution
370	Formerly special status of Jammu and Kashmir (abrogated in 2019)

2.2.4 Parliament and the executive

Office / Body	Key facts
President	Head of state. Elected indirectly by an electoral college. Minimum age 35. Term 5 years. Article 52-62
Vice-President	Ex-officio Chairman of the Rajya Sabha. Minimum age 35
Prime Minister	Real executive head. Appointed by the President; leader of the majority in the Lok Sabha
Lok Sabha	Maximum strength 552. Term 5 years. Minimum age to contest 25. Presided over by the Speaker
Rajya Sabha	Maximum strength 250 (238 elected + 12 nominated). A permanent house - never dissolved; one-third retire every 2 years. Term 6 years. Minimum age 30
Money Bill	Can be introduced only in the Lok Sabha , and only with the President's recommendation. The Rajya Sabha can delay it for at most 14 days
Joint sitting	Held for an ordinary bill deadlock; presided over by the Speaker of the Lok Sabha . Not applicable to Money Bills or Constitutional Amendment Bills
Supreme Court	Established 1950. Currently 34 judges including the CJI. Judges retire at 65

Office / Body	Key facts
High Court judges	Retire at 62
Attorney General	The government's chief legal adviser - Article 76
CAG	Comptroller and Auditor General - Article 148, called the guardian of the public purse
Election Commission	Article 324. A multi-member body
UPSC	Article 315
Finance Commission	Article 280, constituted every 5 years
NITI Aayog	Replaced the Planning Commission on 1 January 2015. The Prime Minister is its Chairperson

Q. [PYQ-TYPE] Who is called the Father of the Indian Constitution?

- (a) Jawaharlal Nehru
- (b) Dr. Rajendra Prasad
- (c) **Dr. B. R. Ambedkar**
- (d) Mahatma Gandhi

Why: Ambedkar chaired the **Drafting Committee**. Note the distinction that examiners exploit: **Dr. Rajendra Prasad** was the President of the **Constituent Assembly**, and later India's first President.

Q. [PYQ-TYPE] Which Article of the Constitution deals with the Right to Education?

- (a) Article 19
- (b) **Article 21A**
- (c) Article 32
- (d) Article 45

Why: **Article 21A**, inserted by the **86th Amendment, 2002**, makes free and compulsory education for ages 6-14 a fundamental right. **Article 45** is the related *Directive Principle* on early childhood care - a deliberate distractor.

Q. [EXPECTED] [TRAP] Which Fundamental Right was removed from the Constitution?

- (a) Right to Equality
- (b) **Right to Property**
- (c) Right to Freedom
- (d) Right against Exploitation

Why: The **Right to Property** was deleted as a Fundamental Right by the **44th Amendment (1978)** and shifted to **Article 300A** as an ordinary legal right. So India now has **six** Fundamental Rights, not seven.

Q. [EXPECTED] The Rajya Sabha is a permanent house because:

- (a) Its members serve for life
- (b) **It is never dissolved; one-third of members retire every two years**
- (c) It has no elections
- (d) The President chairs it

Why: Its continuity is the key contrast with the Lok Sabha, which can be dissolved. Members serve **6-year** terms staggered so a third retire biennially.

2.3 Indian History

2.3.1 Ancient India

Topic	Key facts
Indus Valley Civilization	About 2500-1750 BCE. Bronze Age. Major sites: Harappa (Punjab, Pakistan, on the Ravi), Mohenjodaro (Sindh, on the Indus, means "mound of the dead"), Dholavira and Lothal (Gujarat - Lothal had a dockyard), Kalibangan (Rajasthan), Rakhigarhi (Haryana, the largest Indian site). Known for town planning, grid streets, drainage, the Great Bath at Mohenjodaro. The script is undeciphered
Vedic Period	Four Vedas: Rigveda (oldest, hymns), Samaveda (music), Yajurveda (sacrificial formulae), Atharvaveda (spells and medicine). "Satyameva Jayate" comes from the Mundaka Upanishad
Buddhism	Founded by Gautama Buddha (563-483 BCE), born at Lumbini (Nepal), enlightenment at Bodh Gaya , first sermon at Sarnath , died at Kushinagar . Four Noble Truths, Eightfold Path. Texts in Pali
Jainism	24th Tirthankara Mahavira (599-527 BCE), born at Kundagrama near Vaishali. First Tirthankara Rishabhanatha . Principles: non-violence, truth, non-stealing, celibacy, non-possession
Mauryan Empire	Founded by Chandragupta Maurya (321 BCE) with Chanakya (Kautilya) as adviser, author of the Arthashastra . Ashoka (268-232 BCE) fought the Kalinga war (261 BCE) and embraced Buddhism. Ashoka's edicts are in Prakrit in Brahmi script. The Sarnath Lion Capital is India's national emblem
Gupta Empire	Called the Golden Age of India. Chandragupta I , Samudragupta (the Indian Napoleon), Chandragupta II Vikramaditya . Aryabhata , Kalidasa , Varahamihira , Brahmagupta belong to this era

2.3.2 Medieval India

Dynasty / Event	Key facts
Delhi Sultanate	1206-1526. Five dynasties: Slave (Mamluk) , Khilji , Tughlaq , Sayyid , Lodi
Qutbuddin Aibak	Founder of the Slave dynasty; began the Qutub Minar
Iltutmish	Introduced the silver tanka and copper jital
Razia Sultan	The only woman ruler of the Delhi Sultanate
Alauddin Khilji	Market control reforms; built Siri Fort and Alai Darwaza
Muhammad bin Tughlaq	Shifted the capital to Daulatabad ; introduced token currency - both failed
Firoz Shah Tughlaq	Built Firozabad ; installed Ashokan pillars in Delhi
First Battle of Panipat	1526 - Babur defeated Ibrahim Lodi, founding the Mughal Empire
Second Battle of Panipat	1556 - Akbar's forces defeated Hemu
Third Battle of Panipat	1761 - Marathas defeated by Ahmad Shah Abdali
Akbar	1556-1605. Din-i-Ilahi , abolished jizya, Mansabdari system. Navratnas included Birbal, Tansen, Todar Mal, Abul Fazl
Shah Jahan	Built the Taj Mahal (Agra), the Red Fort and Jama Masjid (Delhi), and moved the capital to Shahjahanabad
Aurangzeb	Last powerful Mughal; reimposed jizya
Battle of Plassey	1757 - Robert Clive defeated Siraj-ud-Daula; the beginning of British political power in India
Battle of Buxar	1764 - established British control over Bengal, Bihar and Odisha

Dynasty / Event	Key facts
Shivaji	Founder of the Maratha empire; crowned at Raigad in 1674
Vijayanagara Empire	Founded 1336 by Harihara and Bukka; Krishnadevaraya its greatest ruler; capital Hampi

2.3.3 Modern India and the freedom struggle

Year	Event
1857	First War of Independence / Sepoy Mutiny. Began at Meerut on 10 May. Mangal Pandey at Barrackpore. Leaders: Rani Lakshmibai, Nana Saheb, Tantia Tope, Bahadur Shah Zafar, Begum Hazrat Mahal
1885	Indian National Congress founded by A. O. Hume . First session in Bombay, presided over by W. C. Bonnerjee
1905	Partition of Bengal by Lord Curzon; Swadeshi Movement began
1906	Muslim League founded at Dhaka
1911	Capital shifted from Calcutta to Delhi ; Partition of Bengal annulled
1915	Gandhi returned to India from South Africa
1917	Champaran Satyagraha - Gandhi's first satyagraha in India
1919	Jallianwala Bagh massacre (13 April, Amritsar, General Dyer); Rowlatt Act
1920	Non-Cooperation Movement launched
1922	Chauri Chaura incident; Gandhi withdrew Non-Cooperation
1927	Simon Commission appointed - "no Indian member", hence the boycott
1929	Lahore Session - Purna Swaraj resolution under Jawaharlal Nehru
1930	Dandi March / Salt Satyagraha (12 March to 6 April); Civil Disobedience Movement
1931	Gandhi-Irwin Pact; Bhagat Singh, Rajguru and Sukhdev executed (23 March)
1935	Government of India Act - provincial autonomy
1942	Quit India Movement - "Do or Die". Also the Cripps Mission
1943	Azad Hind Fauj reorganised by Subhas Chandra Bose - "Give me blood and I will give you freedom"
1946	Cabinet Mission; Constituent Assembly formed; Naval mutiny
1947	Independence, 15 August . Mountbatten the last Viceroy; Radcliffe Line

Famous slogans "Swaraj is my birthright" - **Bal Gangadhar Tilak**. "Do or Die" and "Quit India" - **Gandhi**. "Jai Hind" and "Give me blood..." - **Subhas Chandra Bose**. "Inquilab Zindabad" - **Bhagat Singh**. "Satyameva Jayate" - Mundaka Upanishad. "Jai Jawan Jai Kisan" - **Lal Bahadur Shastri**. "Sare Jahan Se Achha" - **Iqbal**.

Titles **Father of the Nation** - Gandhi. **Iron Man of India** - Sardar Patel. **Netaji** - Subhas Chandra Bose. **Punjab Kesari** - Lala Lajpat Rai. **Lokmanya** - Tilak. **Deshbandhu** - C. R. Das. **Gurudev** - Rabindranath Tagore. **Frontier Gandhi** - Khan Abdul Ghaffar Khan.

Q. [PYQ-TYPE] The Jallianwala Bagh massacre took place in which year?

- (a) 1909
- (b) **1919**
- (c) 1920
- (d) 1922

Why: **13 April 1919** at Amritsar, on Baisakhi day, under General Dyer. It followed the **Rowlatt Act** of the same year and triggered the Non-Cooperation Movement of 1920.

Q. [PYQ-TYPE] Who founded the Indian National Congress?

- (a) Bal Gangadhar Tilak
- (b) **A. O. Hume**
- (c) W. C. Bonnerjee
- (d) Dadabhai Naoroji

Why: **A. O. Hume**, a retired British civil servant, founded it in **1885**. **W. C. Bonnerjee** presided over the first session - the classic distractor pair.

Q. [EXPECTED] The Dandi March was launched to protest against:

- (a) The partition of Bengal
- (b) **The salt tax**
- (c) The Rowlatt Act
- (d) The Simon Commission

Why: Gandhi marched from **Sabarmati Ashram to Dandi** (about 385 km, 12 March to 6 April 1930) to break the **salt law**, launching the Civil Disobedience Movement.

Q. [EXPECTED] [TRAP] Where did the Revolt of 1857 begin?

- (a) Delhi
- (b) **Meerut**
- (c) Kanpur
- (d) Jhansi

Why: It began at **Meerut on 10 May 1857**. **Mangal Pandey's** earlier act was at **Barrackpore** (March 1857) - often confused with the outbreak itself.

2.4 Geography

2.4.1 India - physical geography

Fact	Detail
Area	About 32.87 lakh sq km - 7th largest country
Latitudinal extent	8 degrees 4 min N to 37 degrees 6 min N
Tropic of Cancer passes through	8 states: Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura, Mizoram
Standard Meridian	82 degrees 30 min E , passing through Mirzapur (Uttar Pradesh)
Highest peak in India	Kangchenjunga (8586 m, Sikkim). K2 is in PoK
Longest river	Ganga . Longest entirely within India also Ganga
Largest state by area	Rajasthan ; smallest Goa
Most populous state	Uttar Pradesh
Longest coastline among states	Gujarat
States sharing a border with the most other states	Uttar Pradesh (8)
Land neighbours	7 - Pakistan, Afghanistan, China, Nepal, Bhutan, Myanmar, Bangladesh (longest border: Bangladesh)
Southernmost point of India	Indira Point , Great Nicobar
Local time reference	IST is UTC + 5:30

Important lines and passes	Detail
Radcliffe Line	India - Pakistan
Line of Control (LoC)	India - Pakistan in Jammu and Kashmir
McMahon Line	India - China
Durand Line	Pakistan - Afghanistan
Palk Strait	Separates India from Sri Lanka
Nathu La	Sikkim, to Tibet
Zoji La	Ladakh
Rohtang Pass	Himachal Pradesh
Shipki La	Himachal Pradesh

Rivers	Origin and notes
Ganga	Gangotri glacier as Bhagirathi; joins Alaknanda at Devprayag
Yamuna	Yamunotri; the longest tributary of the Ganga; flows past Delhi
Brahmaputra	Called Tsangpo in Tibet, Jamuna in Bangladesh
Indus	Rises near Mansarovar; the Indus Waters Treaty (1960) allots the Ravi, Beas and Sutlej to India
Godavari	Longest river of peninsular India , called Dakshina Ganga
Narmada and Tapi	Flow west into the Arabian Sea through rift valleys
Krishna, Kaveri, Mahanadi	Flow east into the Bay of Bengal

2.4.2 World geography essentials

Fact	Answer
Largest continent	Asia
Largest ocean	Pacific
Largest desert	Sahara (hot); Antarctica is the largest cold desert
Longest river in the world	Nile (Amazon is the largest by volume)
Highest mountain	Mount Everest (8848.86 m)
Largest country by area	Russia
Deepest ocean point	Mariana Trench
Largest island	Greenland
Highest waterfall	Angel Falls (Venezuela)
Largest freshwater lake by area	Lake Superior; deepest is Lake Baikal
Prime Meridian passes through	Greenwich, London (0 degrees longitude)
Number of continents / oceans	7 continents, 5 oceans

Nickname	Place
Land of the Rising Sun	Japan
Land of the Midnight Sun	Norway
Land of Thunderbolts	Bhutan
Pearl of the Antilles	Cuba
Gift of the Nile	Egypt
Roof of the World	Pamir, Tibet
Playground of Europe	Switzerland
City of Seven Hills	Rome
Emerald Isle	Ireland
Windy City	Chicago
Pink City	Jaipur
Manchester of India	Ahmedabad
Silicon Valley of India	Bengaluru
City of Nawabs	Lucknow
Queen of the Arabian Sea	Kochi

Q. [PYQ-TYPE] The Tropic of Cancer passes through how many Indian states?

- (a) 6
- (b) 7
- (c) **8**
- (d) 9

Why: Eight: Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura and Mizoram. A useful mnemonic in order from west to east: **Gu-Ra-Ma-Chha-Jha-We-Tri-Mi**.

Q. [PYQ-TYPE] Which is the longest river of peninsular India?

- (a) Krishna
- (b) **Godavari**
- (c) Kaveri
- (d) Narmada

Why: The **Godavari**, known as **Dakshina Ganga**. Note that the Narmada and Tapi are the notable **westward-flowing**

peninsular rivers.

Q. [EXPECTED] India's standard meridian passes through which city?

- (a) Delhi
- (b) Nagpur
- (c) **Mirzapur**
- (d) Allahabad

Why: 82 degrees 30 minutes East passes through **Mirzapur** in Uttar Pradesh, which is why IST is UTC + 5:30.

2.5 General Science

2.5.1 Physics

Concept	Key fact
SI base units	metre (length), kilogram (mass), second (time), ampere (current), kelvin (temperature), mole (substance), candela (luminous intensity)
Newton's first law	Law of inertia - a body continues in its state unless acted on by an external force
Newton's second law	$F = ma$
Newton's third law	Every action has an equal and opposite reaction
Speed of light	3×10^8 m/s
Speed of sound in air	About 343 m/s at 20 degrees C. Sound cannot travel in vacuum; light can
Reflection in a plane mirror	Virtual, erect, same size, laterally inverted
Concave mirror	Converging; used in torches, headlights, shaving mirrors
Convex mirror	Diverging; used as rear-view mirrors because it gives a wider field
Total internal reflection	Basis of optical fibre and the sparkle of diamonds; also mirages
Dispersion	Splitting of white light into seven colours - VIBGYOR
Why the sky is blue	Scattering of shorter wavelengths (Rayleigh scattering)
Electric power	$P = VI$; energy in kWh (1 unit = 1 kWh)
Fuse	A safety device that melts to break excess current
Transformer	Changes AC voltage; works on mutual induction. Does not work on DC

2.5.2 Chemistry

Concept	Key fact
Atomic number	Number of protons
Isotopes	Same atomic number, different mass number
pH scale	0 to 14. Below 7 acidic, 7 neutral, above 7 basic
Common acids	Hydrochloric (stomach), sulphuric (king of chemicals), citric (lemon), acetic (vinegar), lactic (curd), formic (ant sting), oxalic (tomato)
Chemical names	Common salt = sodium chloride; baking soda = sodium bicarbonate ; washing soda = sodium carbonate; bleaching powder = calcium hypochlorite; quicklime = calcium oxide; plaster of Paris = calcium sulphate hemihydrate; caustic soda = sodium hydroxide
Alloys	Brass = copper + zinc; Bronze = copper + tin; Steel = iron + carbon; Stainless steel = iron + chromium + nickel; Solder = lead + tin
Noble gases	Helium, Neon, Argon, Krypton, Xenon, Radon - chemically inert
Lightest gas	Hydrogen. Most abundant element in the universe is also hydrogen

Concept	Key fact
Most abundant gas in air	Nitrogen (about 78%), then oxygen (about 21%)
LPG	Mainly butane and propane
Rusting	Requires both oxygen and moisture
Vitamin C	Ascorbic acid
Hardest natural substance	Diamond (an allotrope of carbon, like graphite)

2.5.3 Biology

Concept	Key fact
Cell	The structural and functional unit of life. Discovered by Robert Hooke (1665)
Powerhouse of the cell	Mitochondria (site of respiration, produces ATP)
Suicidal bags of the cell	Lysosomes
Brain of the cell	Nucleus
Photosynthesis	Occurs in chloroplasts using chlorophyll ; converts CO ₂ and water into glucose, releasing oxygen
Blood	Plasma plus RBC, WBC and platelets. RBC carries oxygen via haemoglobin; WBC fights infection ; platelets clot blood
Human blood groups	A, B, AB, O. O negative is the universal donor; AB positive the universal recipient
Normal human body temperature	37 degrees C / 98.6 degrees F
Heart	Four chambers; the largest artery is the aorta
Largest organ	Skin ; largest internal organ / largest gland is the liver
Smallest bone	Stapes (in the ear); longest bone is the femur
Bones in an adult human	206 (a newborn has about 300)
Total teeth in an adult	32
Vitamin deficiencies	A - night blindness; B1 - beri-beri; C - scurvy; D - rickets; K - poor blood clotting; Iron - anaemia; Iodine - goitre
Diseases caused by viruses	Common cold, influenza, measles, polio, rabies, dengue, chickenpox, AIDS , COVID-19
Diseases caused by bacteria	Tuberculosis, cholera, typhoid, tetanus, diphtheria, plague
Diseases caused by protozoa	Malaria (Plasmodium, spread by the female Anopheles mosquito), amoebic dysentery
Insulin	Produced by the pancreas ; its deficiency causes diabetes

Q. [PYQ-TYPE] Which part of the cell is known as the powerhouse of the cell?

- (a) Nucleus
- (b) Ribosome
- (c) **Mitochondria**
- (d) Lysosome

Why: **Mitochondria** carry out cellular respiration and generate **ATP**, the cell's energy currency. Learn the family: nucleus = brain, **lysosome = suicidal bags**, ribosome = protein factory, chloroplast = kitchen (in plants).

Q. [PYQ-TYPE] Malaria is caused by which organism, and spread by which vector?

- (a) Bacteria; housefly
- (b) **Protozoa (Plasmodium); female Anopheles mosquito**
- (c) Virus; Aedes mosquito
- (d) Fungus; sandfly

Why: Malaria is caused by the **protozoan Plasmodium** and transmitted by the **female Anopheles**. Note the contrast: **dengue and chikungunya** are viral and spread by the **Aedes aegypti** mosquito.

Q. [EXPECTED] Which gas is most abundant in the Earth's atmosphere?

- (a) Oxygen
- (b) **Nitrogen**
- (c) Carbon dioxide
- (d) Argon

Why: **Nitrogen at about 78%**, with oxygen about 21% and argon under 1%. Carbon dioxide is only about 0.04% despite its climatic importance.

Q. [EXPECTED] [TRAP] Which mirror is used as a rear-view mirror in vehicles, and why?

- (a) Concave, because it magnifies
- (b) **Convex, because it gives a wider field of view**
- (c) Plane, because it shows true size
- (d) Concave, because it shows a real image

Why: A **convex** mirror diverges light, producing a small, erect virtual image and covering a **wider field of view**. **Concave** mirrors converge light and are used in torches, headlights and shaving mirrors.

2.6 Indian Economy

Topic	Key facts
RBI	Established 1 April 1935 ; nationalised 1949. India's central bank and the sole authority to issue currency except the one-rupee coin and notes , which the Ministry of Finance issues
Monetary policy tools	Repo rate, reverse repo rate, CRR, SLR, open market operations
Inflation types	Demand-pull (demand exceeds supply) and cost-push (rising input costs). Measured by CPI and WPI
GDP vs GNP	GDP counts output produced within the country; GNP adds net factor income from abroad
Fiscal deficit	Total expenditure minus total revenue excluding borrowings
Budget	Presented on 1 February ; the financial year runs 1 April to 31 March. Article 112 calls it the Annual Financial Statement
GST	Introduced 1 July 2017 ; a destination-based indirect tax with slabs. The GST Council is chaired by the Union Finance Minister
Direct vs indirect tax	Direct: income tax, corporate tax. Indirect: GST, customs duty
NITI Aayog	Replaced the Planning Commission on 1 January 2015
First Five Year Plan	1951-56, based on the Harrod-Domar model, focused on agriculture. The Twelfth was the last (2012-17)
Green Revolution	1960s, led by M. S. Swaminathan ; wheat and rice. White Revolution (milk) - Verghese Kurien. Blue Revolution - fish
SEBI	Securities market regulator, statutory since 1992
Poverty line committees	Tendulkar Committee, Rangarajan Committee

Q. [EXPECTED] Which institution issues currency notes in India?

- (a) Ministry of Finance
- (b) **Reserve Bank of India**
- (c) State Bank of India
- (d) NITI Aayog

Why: The **RBI** issues all currency notes **except the one-rupee note and coins**, which are issued by the Government of India (Ministry of Finance). That exception is exactly what such questions test.

Q. [EXPECTED] GST was implemented in India from:

- (a) 1 April 2017
- (b) **1 July 2017**
- (c) 1 January 2017
- (d) 15 August 2017

Why: **1 July 2017**, now marked as GST Day. It replaced a patchwork of indirect taxes with a single destination-based tax.

2.7 Static Miscellany

2.7.1 Awards and honours

Award	Field
Bharat Ratna	Highest civilian award. Instituted 1954
Padma Vibhushan, Padma Bhushan, Padma Shri	Civilian awards in descending order after Bharat Ratna
Param Vir Chakra	Highest wartime military honour
Ashoka Chakra	Highest peacetime military honour
Major Dhyan Chand Khel Ratna	Highest sporting honour (formerly Rajiv Gandhi Khel Ratna)
Arjuna Award	Outstanding sportspersons
Dronacharya Award	Coaches
Dhyan Chand Award	Lifetime contribution to sport
Jnanpith Award	Literature (Indian languages)
Sahitya Akademi Award	Literature
Dadasaheb Phalke Award	Cinema - highest Indian film honour
Nobel Prize categories	Physics, Chemistry, Medicine, Literature, Peace, Economics (Economics added 1969, awarded by Sweden's central bank)
Ramon Magsaysay Award	Asia's Nobel

2.7.2 Important days

Date	Day
26 January	Republic Day
28 February	National Science Day
8 March	International Women's Day
22 March	World Water Day
7 April	World Health Day
22 April	Earth Day
1 May	Labour Day
5 June	World Environment Day
21 June	International Yoga Day
1 July	National Doctors' Day (India)
15 August	Independence Day
29 August	National Sports Day (Dhyan Chand's birthday)
5 September	Teachers' Day (Dr. S. Radhakrishnan's birthday)
14 September	Hindi Diwas
2 October	Gandhi Jayanti / International Day of Non-Violence
24 October	United Nations Day
14 November	Children's Day (Nehru's birthday)
26 November	Constitution Day
1 December	World AIDS Day

Date	Day
10 December	Human Rights Day

2.7.3 Organisations and headquarters

Organisation	Headquarters
United Nations	New York
UNESCO	Paris
WHO	Geneva
ILO	Geneva
WTO	Geneva
IMF and World Bank	Washington DC
International Court of Justice	The Hague
UNICEF	New York
Interpol	Lyon, France
OPEC	Vienna
SAARC	Kathmandu
ASEAN	Jakarta
NATO	Brussels
Amnesty International	London
Red Cross	Geneva
Asian Development Bank	Manila
BRICS New Development Bank	Shanghai

2.7.4 Sports basics

Sport	Facts
Cricket	11 players. Terms: LBW, googly, maiden over. India won the World Cup in 1983, 2011 (ODI) and 2007, 2024 (T20)
Hockey	India's most successful Olympic sport (8 gold). Major Dhyan Chand is the Wizard of Hockey
Football	11 players. FIFA World Cup every 4 years
Olympics	Every 4 years. Five rings represent five continents. Motto: Citius, Altius, Fortius (Faster, Higher, Stronger) - "Communiter" added in 2021
Trophies - Cricket	Ranji Trophy, Duleep Trophy, Irani Cup, Deodhar Trophy
Trophies - Football	Durand Cup, Santosh Trophy
Trophies - Hockey	Agha Khan Cup, Beighton Cup
Trophies - Badminton	Thomas Cup (men), Uber Cup (women)
Trophies - Tennis	Davis Cup (men), Wimbledon

2.8 Delhi-Specific General Knowledge

2.8.1 Why this chapter exists

DSSSB recruits for the **Government of the National Capital Territory of Delhi**. Questions about Delhi - its administration, history, monuments and civic bodies - appear in DSSSB papers in a way they never do in a national exam. Most candidates never prepare this, which makes it cheap marks.

Fact	Detail
Status	National Capital Territory (NCT) of Delhi - a Union Territory with a legislature, under the 69th Constitutional Amendment Act, 1991
Governing Articles	239AA and 239AB
Executive head	The Lieutenant Governor (LG) , appointed by the President
Head of government	The Chief Minister , leading the Council of Ministers
Legislative Assembly	70 seats; term 5 years
Lok Sabha seats from Delhi	7
Rajya Sabha seats from Delhi	3
Subjects outside the Assembly's power	Public order, police and land - these rest with the Union Government
High Court	Delhi High Court, established 1966
Municipal body	Municipal Corporation of Delhi (MCD) - the three corporations were reunified into a single MCD in 2022 . Also the New Delhi Municipal Council (NDMC) and the Delhi Cantonment Board
Districts	11 revenue districts

2.8.2 Delhi's history and monuments

Monument / Site	Facts
Qutub Minar	Begun by Qutbuddin Aibak , completed by Iltutmish . Tallest brick minaret in the world. UNESCO World Heritage Site . The Iron Pillar in its complex is rust-resistant
Red Fort (Lal Qila)	Built by Shah Jahan (1638-1648) at Shahjahanabad. UNESCO site . The Prime Minister addresses the nation from here on Independence Day
Humayun's Tomb	Built by Humayun's widow Bega Begum ; the first garden-tomb in India and an architectural precursor to the Taj Mahal. UNESCO site
Jama Masjid	Built by Shah Jahan; one of India's largest mosques
India Gate	War memorial by Edwin Lutyens , honouring soldiers of the First World War. The National War Memorial was inaugurated nearby in 2019
Rashtrapati Bhavan	Designed by Edwin Lutyens ; the President's residence
Parliament House	Designed by Herbert Baker . The new Parliament building was inaugurated in 2023
Lotus Temple	Bahai House of Worship
Akshardham	Swaminarayan temple complex

Monument / Site	Facts
Purana Qila	Associated with Humayun and Sher Shah Suri
Jantar Mantar	Astronomical observatory built by Sawai Jai Singh II
Raj Ghat	Memorial to Mahatma Gandhi

The seven cities of Delhi Traditionally: **Qila Rai Pithora** (Tomars/Chauhans), **Siri** (Alauddin Khilji), **Tughlaqabad** (Ghiyasuddin Tughlaq), **Jahanpanah** (Muhammad bin Tughlaq), **Firozabad** (Firoz Shah Tughlaq), **Dinpanah/Purana Qila** (Humayun and Sher Shah), and **Shahjahanabad** (Shah Jahan).

Capital shifted to Delhi Announced at the **Delhi Durbar of 1911**; New Delhi formally inaugurated in **1931**.

Delhi Metro Operated by **DMRC**, established 1995; first line opened in **2002** (Shahdara to Tis Hazari). E. Sreedharan is called the **Metro Man of India**.

Rivers and geography The **Yamuna** flows through Delhi; the **Aravalli** range's northern extension (the Delhi Ridge) runs through it.

Delhi's UNESCO World Heritage Sites **Qutub Minar complex, Red Fort complex, Humayun's Tomb.**

Q. [PYQ-TYPE] Under which Constitutional Amendment did Delhi get the status of National Capital Territory?

- (a) 42nd Amendment
- (b) **69th Amendment**
- (c) 73rd Amendment
- (d) 74th Amendment

Why: The **69th Amendment Act, 1991** inserted **Articles 239AA and 239AB**, creating the NCT of Delhi with a Legislative Assembly and Council of Ministers. Note the distractors: the **73rd** is panchayats and the **74th** is municipalities.

Q. [PYQ-TYPE] Which three subjects are outside the legislative competence of the Delhi Legislative Assembly?

- (a) Health, education, transport
- (b) **Public order, police and land**
- (c) Water, electricity, housing
- (d) Finance, taxation, excise

Why: **Public order, police and land** remain with the Union Government - the defining limitation of Delhi's special status and a very likely DSSSB question.

Q. [EXPECTED] Who designed Rashtrapati Bhavan?

- (a) Herbert Baker
- (b) **Edwin Lutyens**
- (c) Le Corbusier
- (d) Charles Correa

Why: **Edwin Lutyens** designed Rashtrapati Bhavan and India Gate; **Herbert Baker** designed the Secretariat blocks and the old Parliament House. The pair is a standard distractor set.

Q. [EXPECTED] How many Lok Sabha seats does Delhi have?

- (a) 5
- (b) 6
- (c) **7**
- (d) 10

Why: **7 Lok Sabha** seats and **3 Rajya Sabha** seats, with **70** Assembly constituencies.

2.9 Static GK Revision Sheet

2.9.1 The forty facts most likely to appear

Question	Answer
Father of the Indian Constitution	Dr. B. R. Ambedkar
Constitution adopted / came into force	26 Nov 1949 / 26 Jan 1950
Heart and soul of the Constitution	Article 32
Right to Education Article	21A (86th Amendment, 2002)
Fundamental Duties added by	42nd Amendment, 1976
Right removed from Fundamental Rights	Right to Property (44th Amendment)
Directive Principles borrowed from	Ireland
Money Bill can be introduced in	Lok Sabha only
Rajya Sabha maximum strength	250 (238 + 12 nominated)
Supreme Court / High Court retirement age	65 / 62
Guardian of the public purse	CAG (Article 148)
Largest Indus Valley site in India	Rakhigarhi
Ashoka's Kalinga war	261 BCE
Golden Age of India	Gupta period
First Battle of Panipat	1526, Babur vs Ibrahim Lodi
Battle of Plassey	1757
INC founded	1885 by A. O. Hume
Jallianwala Bagh	13 April 1919, Amritsar
Dandi March	12 March 1930, against the salt tax
Quit India Movement	1942, "Do or Die"
Tropic of Cancer through Indian states	8
Indian standard meridian	82 deg 30 min E, Mirzapur
Longest peninsular river	Godavari
Highest peak in India	Kangchenjunga
India's longest land border with	Bangladesh
Powerhouse of the cell	Mitochondria
Suicidal bags of the cell	Lysosomes
Universal donor blood group	O negative
Bones in an adult human	206
Malaria vector	Female Anopheles mosquito
Dengue vector	Aedes aegypti
Most abundant atmospheric gas	Nitrogen (78%)
King of chemicals	Sulphuric acid
Brass / Bronze composition	Copper + zinc / Copper + tin
RBI established	1 April 1935
GST implemented	1 July 2017

Question	Answer
NITI Aayog replaced Planning Commission	1 January 2015
Delhi NCT status via	69th Amendment, 1991
Subjects outside Delhi Assembly's power	Public order, police, land
Delhi's UNESCO sites	Qutub Minar, Red Fort, Humayun's Tomb
Delhi Lok Sabha / Assembly seats	7 / 70
Metro Man of India	E. Sreedharan

PART 3

**GENERAL AWARENESS - CURRENT
AFFAIRS**

5 to 8 marks of the General Awareness section. This is the one Part of this book that cannot be complete, and this chapter explains exactly why and what to do about it.

3.1 The Honest Chapter - Read This First

3.1.1 Why no book can give you current affairs

You asked me not to skip anything and to give this my best. The most useful thing I can do here is be straight with you rather than pad this Part with a list that looks impressive and expires.

Here is the problem in one sentence: **a book is written at one moment, and your exam happens at another.**

Every other Part of this book will still be correct in five years. Pythagoras will not change. Article 21A will not change. This Part starts decaying the day it is written.

Three consequences you need to accept:

- 1. I cannot predict specific current-affairs questions.** Anyone who claims to be guessing and dressing it up. What I *can* do is show you the **categories** that recur every single year, and those categories are genuinely stable.
- 2. Some facts in this Part will already be out of date when you read it.** Office holders change. Tournaments get won by someone else. Where I give a specific name or figure, I have flagged how to verify it.
- 3. A daily habit will beat this chapter.** Fifteen minutes a day for six months is worth more than this entire Part, because it covers the actual window your paper draws from.

What I did and did not verify

I searched for current affairs material while writing this and found the standard aggregator sources, monthly capsules and summit round-ups. But I want to be precise about the limits of that:

I have not independently verified individual current-affairs facts against primary government sources, and current-affairs aggregators contain errors. So rather than transcribing a list of names and dates that I cannot stand behind and that you would be memorising on trust, I have built this Part as a **framework plus a verification method**.

Where I do name something specific, I say where to check it. Where I am not confident, I say so instead of filling the gap. A wrong fact learned confidently is worse than a gap you know you have - in an exam with negative marking, false confidence actively costs you marks.

Sources consulted for the framework and category structure: [GKToday India current affairs](https://www.gktoday.in/current-affairs/category/india-current-affairs/), [Vajiram and Ravi - summits and conferences](https://vajiramandravi.com/current-affairs/important-summits-and-conferences/), [Testbook daily current affairs](https://testbook.com/current-affairs), [PW last six months current affairs](https://pw.live/railway/exams/last-6-months-current-affairs-2026-october-march-2026-by-raja-gupta-sir). Content from these sources was paraphrased and condensed for compliance with licensing restrictions.

3.1.2 What the examiner actually asks

This is the part that *is* predictable, and it is worth more than a list of facts. Across DSSSB and similar papers, current-affairs questions cluster into a small number of shapes.

Category	Typical question	Weight
Appointments	Who was appointed CJI / RBI Governor / Chief Election Commissioner / Army Chief?	High
Government schemes	Which ministry launched scheme X? What is its objective?	High
Sports	Who won a major tournament? Where were the last Olympics / Asian Games held? Where are the next?	High
Awards	Who received the Bharat Ratna / Jnanpith / Dadasaheb Phalke / Nobel this year?	High
Summits and conferences	Which country hosted the G20 / BRICS / SCO / COP summit? What was its theme?	Medium
Reports and indices	Which organisation publishes the Human Development Index / Global Hunger Index / Ease of Doing Business?	Medium

Category	Typical question	Weight
Defence	Which missile was test-fired? Which warship was commissioned? What was the exercise called?	Medium
Space and science	Recent ISRO missions and their objectives	Medium
Books and authors	Who wrote a recently released notable book?	Low
Obituaries	Notable deaths and what the person was known for	Low
Days and themes	The theme of a major international day this year	Low
State news	New capitals, new districts, state schemes - especially Delhi	Medium for DSSSB

The single most valuable insight in this Part

Notice that **most of those categories are questions about a slot, not about an event**. "Who is the RBI Governor" is a slot that always exists; only the occupant changes. "Which country hosted the G20" is a slot; only the host changes.

So you do not need to read everything. You need to **maintain a small table of slots** and keep the occupants current. That table is perhaps 60 lines long, and it covers the majority of appointment-and-position questions.

I have given you the table structure in Chapter 3.2. Filling it in is a 30-minute job, and updating it is a 5-minute monthly job.

3.2 The Slot Table - Build This Yourself

3.2.1 How to use this chapter

Below are the slots that recur. **Copy this into a notebook and fill in the current occupant**, then update it monthly. I have deliberately left the answers blank rather than fill them with values that may be stale by the time you read this - a blank you fill in is reliable; a printed name you trust may not be.

3.2.2 Constitutional and government offices

Slot	Fill in	Where to verify
President of India	—	presidentofindia.nic.in
Vice-President	—	Official Rajya Sabha site
Prime Minister	—	pmindia.gov.in
Chief Justice of India	—	sci.gov.in
Speaker of the Lok Sabha	—	sansad.in
Chief Election Commissioner	—	eci.gov.in
RBI Governor	—	rbi.org.in
CAG of India	—	cag.gov.in
Attorney General of India	—	Ministry of Law
Chairman, NITI Aayog / Vice-Chairman	—	niti.gov.in
Cabinet Secretary	—	cabsec.gov.in
Chief of Defence Staff	—	mod.gov.in
Chiefs of Army / Navy / Air Staff	—	mod.gov.in
Chairman, UPSC	—	upsc.gov.in
Chairman, ISRO	—	isro.gov.in
Chairman, SEBI	—	sebi.gov.in
Solicitor General	—	Ministry of Law

3.2.3 Delhi-specific slots - important for DSSSB

Slot	Fill in	Where to verify
Lieutenant Governor of Delhi	—	ltgovdelhi.nic.in
Chief Minister of Delhi	—	delhi.gov.in
Education Minister of Delhi	—	edudel.nic.in
Chief Secretary of Delhi	—	delhi.gov.in
Chief Justice, Delhi High Court	—	delhihighcourt.nic.in
Mayor of Delhi (MCD)	—	mcdonline.nic.in
Delhi Police Commissioner	—	delhipolice.gov.in
Recent major Delhi government schemes	—	delhi.gov.in

Why the Delhi slots matter more than you think

DSSSB is a **Delhi government** recruitment board. Its General Awareness papers have historically included questions about the Delhi administration, Delhi schemes and Delhi civic bodies that a candidate preparing from a national current-affairs magazine would simply miss.

This is the cheapest differentiator in the whole General Awareness section. The Delhi government's own website carries its scheme announcements, and the Education Directorate site carries education-specific ones - which are doubly relevant if you are applying for a teaching post.

Spend twenty minutes a month on Delhi-specific news. Almost none of your competition will.

3.2.4 International slots

Slot	Fill in
Secretary-General of the UN	_____
Director-General of WHO	_____
Managing Director of the IMF	_____
President of the World Bank	_____
Director-General of the WTO	_____
Secretary-General of SAARC	_____
Current G20 host / next host	_____
Current BRICS chair	_____
Latest COP (climate summit) host and number	_____
Latest Olympic host / next Olympic host	_____
Latest FIFA World Cup winner	_____
Latest Cricket World Cup / T20 World Cup winner	_____

3.2.5 Reports and indices - which body publishes what

This table is **static** - the publisher does not change even though India's rank does. Learn the publisher, and update only the rank.

Report / Index	Published by
Human Development Index (HDI)	UNDP
World Development Report	World Bank
Ease of Doing Business (historically)	World Bank
Global Hunger Index	Concern Worldwide and Welthungerhilfe
Global Competitiveness Report	World Economic Forum
Global Gender Gap Report	World Economic Forum
World Happiness Report	UN Sustainable Development Solutions Network
Corruption Perceptions Index	Transparency International
Press Freedom Index	Reporters Without Borders
Global Innovation Index	WIPO
World Economic Outlook	IMF
State of the World's Children	UNICEF
Global Peace Index	Institute for Economics and Peace
Environmental Performance Index	Yale and Columbia Universities
India State of Forest Report	Forest Survey of India
Economic Survey of India	Ministry of Finance
NITI Aayog SDG India Index	NITI Aayog
ASER (education report)	Pratham

Q. [PYQ-TYPE] The Human Development Index is published by which organisation?

- (a) World Bank
- (b) IMF
- (c) **UNDP**
- (d) WEF

Why: The **UNDP** publishes the HDI, which combines life expectancy, education and per capita income. Note the frequent distractors: the **World Bank** publishes the World Development Report, the **WEF** publishes the Global Competitiveness and Gender Gap reports, and the **IMF** publishes the World Economic Outlook. The **publisher** is the stable, examinable fact; the rank changes yearly.

Q. [EXPECTED] The Global Hunger Index is jointly published by:

- (a) UNDP and WHO
- (b) **Concern Worldwide and Welthungerhilfe**
- (c) FAO and WFP
- (d) World Bank and IMF

Why: This one is commonly misattributed to a UN body. It is published by two NGOs - **Concern Worldwide** (Ireland) and **Welthungerhilfe** (Germany).

3.3 Government Schemes - The Framework

3.3.1 The schemes that keep appearing

Scheme questions are usually of the form "which ministry runs X?" or "what is the objective of X?" The **major flagship schemes are stable over years**, so these are learnable. Newer schemes must come from your daily reading.

Scheme	Objective / Ministry
Swachh Bharat Mission	Sanitation and cleanliness. Launched 2 October 2014
Beti Bachao Beti Padhao	Child sex ratio and girls' education. Ministry of Women and Child Development
Pradhan Mantri Jan Dhan Yojana	Financial inclusion, zero-balance bank accounts (2014)
Ayushman Bharat / PM-JAY	Health insurance cover for poor families. Ministry of Health
PM Awas Yojana	Housing for all
Ujjwala Yojana (PMUY)	LPG connections to poor households
Make in India	Manufacturing and investment
Digital India	Digital infrastructure and services
Skill India / PMKVY	Skill development. Ministry of Skill Development
Atal Pension Yojana	Pension for the unorganised sector
Sukanya Samriddhi Yojana	Savings scheme for a girl child
PM Kisan Samman Nidhi	Income support to farmers
Mid-Day Meal / PM POSHAN	Nutrition for school children. Ministry of Education
Samagra Shiksha	Integrated school education scheme
Sarva Shiksha Abhiyan	Universal elementary education (2000-01)
National Education Policy 2020	5+3+3+4 structure; replaced NPE 1986
Atmanirbhar Bharat	Self-reliant India economic package
Jal Jeevan Mission	Tap water to every rural household
Fit India Movement	Fitness and sport
Startup India	Entrepreneurship support

The education-scheme angle for teaching posts

If you are applying for TGT, PGT or PRT, pay special attention to **education** schemes - Samagra Shiksha, PM POSHAN, NEP 2020, NIPUN Bharat (foundational literacy and numeracy), and Delhi's own education initiatives. These sit at the intersection of General Awareness and Teaching Methodology, so a single fact can earn you marks in **both** sections.

Q. [PYQ-TYPE] The Swachh Bharat Mission was launched on:

- (a) 15 August 2014
- (b) **2 October 2014**
- (c) 26 January 2015
- (d) 2 October 2015

Why: Launched on **2 October 2014**, Gandhi's birth anniversary, which is the mnemonic - a cleanliness mission launched on Gandhi Jayanti.

Q. [EXPECTED] PM POSHAN is the new name of which scheme?

- (a) Ayushman Bharat
- (b) **Mid-Day Meal Scheme**
- (c) Jan Dhan Yojana
- (d) Ujjwala Yojana

Why: The **Mid-Day Meal Scheme** was renamed **PM POSHAN** (Poshan Shakti Nirman). Do not confuse it with **ICDS**, which serves children **under 6** through Anganwadis, whereas PM POSHAN serves **school-going** children.

3.4 Your Daily Current Affairs System

3.4.1 The 15-minute routine

THE DAILY ROUTINE - 15 minutes, every day, no exceptions

MINUTES 1-8 : Read one reliable daily current-affairs digest.
Do NOT read a newspaper cover to cover - it is not an efficient use of exam preparation time.

MINUTES 9-13 : Write into your notebook ONLY items that fit a slot or a category from Chapter 3.1.2:
appointment / scheme / sport / award / summit / report / defence / space / Delhi
One line each. Date on the left.
If an item fits no category, SKIP IT.

MINUTES 14-15 : Re-read yesterday's lines and last Sunday's lines.
This is the spaced repetition that makes it stick.

WEEKLY (20 min) : Re-read the whole week. Update your slot table.

MONTHLY (40 min) : Re-read the month. Attempt a monthly current-affairs quiz to find gaps.

FINAL 30 DAYS : Read ONLY your own notebook, plus a last-6-months compilation. Do not start new sources at this stage.

3.4.2 What to read

Type of source	Why
A daily current-affairs digest aimed at competitive exams	Pre-filtered for exam relevance; saves hours over reading a full newspaper
One monthly current-affairs compilation	Catches what you missed and consolidates
PIB (pib.gov.in) press releases	The primary source for government schemes and appointments. If a digest and PIB disagree, PIB is right
delhi.gov.in and edudel.nic.in	Delhi-specific news that national sources omit - your differentiator for DSSSB
A last-6-months compilation, in the final month	The highest-yield single revision document before the exam

The mistake that costs candidates the most marks

The commonest failure is **starting current affairs two weeks before the exam**. It cannot work, because current-affairs questions are drawn from a window of roughly six to twelve months. Two weeks of cramming covers perhaps 5% of that window.

The second commonest failure is the opposite: **reading everything and retaining nothing**. Candidates spend an hour daily on newspapers and remember none of it, because they never wrote anything down and never revised.

The routine above fixes both. It is short enough to sustain for months and structured enough to retain. **Fifteen disciplined minutes daily beats two undisciplined hours.**

3.4.3 A realistic expectation for this section

Your current-affairs preparation	Realistic score out of the 5-8 CA questions
No preparation at all	1-2 (pure luck)
Two weeks of cramming before the exam	2-3
Three months of the daily routine	4-5

Your current-affairs preparation	Realistic score out of the 5-8 CA questions
Six months or more of the daily routine, with the slot table maintained	5-7

Combine that with the static GK in Part 2 - which is fully learnable and worth 12-15 of the 20 - and you can see how the section adds up:

Static GK done properly (12-15) + current affairs done properly (5-7) = 17-20 out of 20. Static GK alone caps you at about 15. That is precisely why this Part exists, and why it cannot be skipped.

3.5 Current Affairs Practice - Question Patterns

3.5.1 Questions built on the stable parts

The questions below test the **structural** knowledge that does not expire - publishers, ministries, mandates, host-rotation patterns. These are the current-affairs marks you can actually secure in advance.

Q. [PYQ-TYPE] Which organisation publishes the World Economic Outlook?

- (a) World Bank
- (b) **IMF**
- (c) WEF
- (d) UNCTAD

Why: The **IMF** publishes the World Economic Outlook twice yearly. Contrast the **World Bank's** World Development Report. Publisher-to-report matching is the most reliably examinable current-affairs knowledge.

Q. [EXPECTED] The Economic Survey of India is presented by which ministry, and when?

- (a) Ministry of Commerce, in March
- (b) **Ministry of Finance, a day before the Union Budget**
- (c) NITI Aayog, in April
- (d) RBI, in January

Why: The **Ministry of Finance** (prepared by the Chief Economic Adviser) tables it usually the day before the Budget, which is presented on **1 February**.

Q. [EXPECTED] The Nobel Prize in which field is NOT awarded by a Swedish institution?

- (a) Physics
- (b) Chemistry
- (c) Literature
- (d) **Peace**

Why: The **Nobel Peace Prize** is awarded by the **Norwegian Nobel Committee in Oslo**; all others are awarded in Sweden. Also note the **Economics** prize was added in **1969** and is funded by Sweden's central bank, not by Nobel's original will.

Q. [EXPECTED] The Bharat Ratna was instituted in which year?

- (a) 1950
- (b) **1954**
- (c) 1947
- (d) 1960

Why: **1954**, alongside the Padma awards. It is India's highest civilian honour and can be awarded posthumously. A maximum of three may be conferred in a year.

Q. [EXPECTED] Which body conducts the census of India?

- (a) NITI Aayog
- (b) **Office of the Registrar General and Census Commissioner, Ministry of Home Affairs**
- (c) Ministry of Statistics
- (d) Election Commission

Why: Conducted by the **Registrar General and Census Commissioner** under the **Ministry of Home Affairs**. India's first complete census was **1881**, and it has been decennial since.

Q. [EXPECTED] The theme-based international days are announced by which body?

- (a) Individual national governments
- (b) **The United Nations and its specialised agencies**
- (c) WEF
- (d) BRICS

Why: The **UN** and agencies such as WHO and UNESCO designate observance days and set annual themes. **The day itself is static and learnable** (5 June is World Environment Day); only the **theme** changes annually - so learn the dates now and pick up the themes from your daily reading.

PART 4

ARITHMETICAL AND NUMERICAL ABILITY

20 marks. Class 10 level, a closed topic list, and entirely technique. The only real constraint is speed. Target 18-20.

4.1 The Master Formula Sheet

4.1.1 Read this chapter first, and revise it weekly

Everything in Part 4 rests on these formulas. If you learn nothing else from this Part, learn this chapter. Most candidates lose maths marks not because a question was hard but because they could not recall the formula in the 60 seconds available.

[FORMULA] PERCENTAGE

$$\begin{aligned} x\% \text{ of } y &= (x/100) \times y \\ x \text{ as a \% of } y &= (x/y) \times 100 \\ \text{Percentage change} &= (\text{change} / \text{ORIGINAL value}) \times 100 \end{aligned}$$

If a value increases by $x\%$ then decreases by $x\%$,
the NET result is always a DECREASE of $(x^2/100)\%$.

Successive changes of $a\%$ and $b\%$:
net % change = $a + b + (ab/100)$ (use signs: decrease is negative)

If A is $x\%$ more than B, then B is $[100x/(100+x)]\%$ less than A.
If A is $x\%$ less than B, then B is $[100x/(100-x)]\%$ more than A.

[FORMULA] PROFIT AND LOSS

$$\begin{aligned} \text{Profit} &= \text{SP} - \text{CP} \\ \text{Loss} &= \text{CP} - \text{SP} \\ \text{Profit \%} &= (\text{Profit} / \text{CP}) \times 100 &<- \text{always on COST PRICE} \\ \text{Loss \%} &= (\text{Loss} / \text{CP}) \times 100 \end{aligned}$$

$$\begin{aligned} \text{SP} &= \text{CP} \times (100 + \text{Profit}\%) / 100 \\ \text{CP} &= \text{SP} \times 100 / (100 + \text{Profit}\%) \end{aligned}$$

$$\begin{aligned} \text{Discount} &= \text{MP} - \text{SP} \\ \text{Discount \%} &= (\text{Discount} / \text{MP}) \times 100 &<- \text{always on MARKED PRICE} \end{aligned}$$

Successive discounts of $a\%$ and $b\%$:
single equivalent discount = $a + b - (ab/100)$

If an article is sold at two different prices with the SAME
percentage profit and loss, there is always a NET LOSS of $(x^2/100)\%$.

[FORMULA] SIMPLE AND COMPOUND INTEREST

$$\begin{aligned} \text{SI} &= (P \times R \times T) / 100 \\ \text{Amount} &= P + \text{SI} \end{aligned}$$

$$\begin{aligned} \text{CI: Amount } A &= P (1 + R/100)^T \\ \text{CI} &= A - P \end{aligned}$$

Compounded half-yearly: rate becomes $R/2$, time becomes $2T$
Compounded quarterly : rate becomes $R/4$, time becomes $4T$

$$\begin{aligned} \text{For 2 years: } \text{CI} - \text{SI} &= P (R/100)^2 \\ \text{For 3 years: } \text{CI} - \text{SI} &= P (R/100)^2 \times (300 + R) / 100 \end{aligned}$$

Money DOUBLES in $100/R$ years at simple interest.

[FORMULA] RATIO AND PROPORTION

$a : b = c : d$ means $ad = bc$ (product of extremes = product of means)

If $a:b = c:d$, then:
componendo $(a+b):b = (c+d):d$

dividendo $(a-b):b = (c-d):d$

To divide amount X in ratio a:b:c ->

shares are $Xa/(a+b+c)$, $Xb/(a+b+c)$, $Xc/(a+b+c)$

Mean proportional between a and b = $\sqrt{ab}$

Third proportional to a and b = b^2 / a

Fourth proportional to a, b, c = bc / a

[FORMULA] AVERAGE

Average = (sum of all observations) / (number of observations)

Sum of first n natural numbers = $n(n+1)/2$

Average of first n natural numbers = $(n+1)/2$

Sum of squares of first n numbers = $n(n+1)(2n+1)/6$

Sum of cubes of first n numbers = $[n(n+1)/2]^2$

Average of first n odd numbers = n

Average of first n even numbers = $n + 1$

If the average of n numbers is A and one number x is replaced by y,
the new average = $A + (y - x)/n$

[FORMULA] TIME AND WORK

If A does a job in 'a' days, A's one-day work = $1/a$

A and B together take $ab/(a+b)$ days

A, B, C together take $1 / (1/a + 1/b + 1/c)$ days

If A is twice as fast as B, then A:B time ratio = 1:2
and A:B work ratio = 2:1

PIPES AND CISTERNS: an inlet is POSITIVE work, an outlet is NEGATIVE.
Filled together in $1 / (1/a - 1/b)$ when b is the leak.

$M_1 D_1 H_1 / W_1 = M_2 D_2 H_2 / W_2$
(men x days x hours per day, divided by work done)

[FORMULA] TIME, SPEED AND DISTANCE

Speed = Distance / Time

UNIT CONVERSION (memorise these two):

km/hr to m/s : multiply by $5/18$

m/s to km/hr : multiply by $18/5$

Average speed for equal DISTANCES at speeds x and y = $2xy/(x+y)$
(this is the harmonic mean - NOT the simple average)

Average speed for equal TIMES at speeds x and y = $(x+y)/2$

TRAINS

Train passing a POLE : distance = length of train

Train passing a PLATFORM: distance = train length + platform length

Same direction (overtaking): relative speed = difference of speeds

Opposite direction (crossing): relative speed = sum of speeds

BOATS AND STREAMS

Downstream speed = boat speed + stream speed

Upstream speed = boat speed - stream speed

Boat speed = $(\text{down} + \text{up}) / 2$

Stream speed = $(\text{down} - \text{up}) / 2$

[FORMULA] MENSURATION - 2D

Square	: area = a^2	perimeter = $4a$	diagonal = $a \times \sqrt{2}$
Rectangle	: area = $l \times b$	perimeter = $2(l+b)$	diagonal = $\sqrt{l^2+b^2}$
Triangle	: area = $(1/2) \times \text{base} \times \text{height}$		
Equilateral	: area = $(\sqrt{3}/4) a^2$	height = $(\sqrt{3}/2) a$	
Circle	: area = πr^2	circumference = $2 \pi r$	
Parallelogram	: area = base $\times$ height		
Rhombus	: area = $(1/2) d_1 \times d_2$		
Trapezium	: area = $(1/2)(\text{sum of parallel sides}) \times \text{height}$		
Sector	: area = $(\theta/360) \pi r^2$		

[FORMULA] MENSURATION - 3D

Cube	: volume = a^3	total surface = $6a^2$	diagonal = $a \times \sqrt{3}$
Cuboid	: volume = lbh	total surface = $2(lb+bh+hl)$	
		diagonal = $\sqrt{l^2+b^2+h^2}$	
Cylinder	: volume = $\pi r^2 h$	curved surface = $2 \pi r h$	
		total surface = $2 \pi r (r + h)$	
Cone	: volume = $(1/3) \pi r^2 h$	slant $l = \sqrt{r^2 + h^2}$	
		curved surface = $\pi r l$	total = $\pi r (r + l)$
Sphere	: volume = $(4/3) \pi r^3$	surface = $4 \pi r^2$	
Hemisphere	: volume = $(2/3) \pi r^3$	total surface = $3 \pi r^2$	

[FORMULA] NUMBER SYSTEM

Dividend = Divisor $\times$ Quotient + Remainder

HCF $\times$ LCM = product of the two numbers

HCF of fractions = HCF of numerators / LCM of denominators

LCM of fractions = LCM of numerators / HCF of denominators

DIVISIBILITY RULES

- 2 : last digit is even
- 3 : sum of digits divisible by 3
- 4 : last TWO digits divisible by 4
- 5 : last digit is 0 or 5
- 6 : divisible by both 2 and 3
- 8 : last THREE digits divisible by 8
- 9 : sum of digits divisible by 9
- 11 : (sum of odd-position digits) - (sum of even-position digits)
is 0 or divisible by 11

4.2 Number System, LCM and HCF

4.2.1 Questions

Q. [PYQ-TYPE] [NUMERICAL] The LCM of two numbers is 495 and their HCF is 5. If their sum is 100, find their difference.

- (a) 10
- (b) 10
- (c) 46
- (d) 54

Why: Use **HCF x LCM = product**: $5 \times 495 = 2475$. So the two numbers have sum 100 and product 2475. Solving $x(100-x) = 2475$ gives $x^2 - 100x + 2475 = 0$, which factorises as $(x-45)(x-55) = 0$. The numbers are 45 and 55, so the difference is 10.

Q. [PYQ-TYPE] [NUMERICAL] Find the largest number that divides 62, 132 and 237 leaving the same remainder in each case.

- (a) 30
- (b) 35
- (c) 25
- (d) 40

Why: Method: when the remainder is the same but unknown, take the **differences** and find their HCF. Differences: $132-62 = 70$, $237-132 = 105$, $237-62 = 175$. HCF of 70, 105 and 175 is 35. (Check: $62 = 35 \times 1 + 27$, $132 = 35 \times 3 + 27$, $237 = 35 \times 6 + 27$. Same remainder 27.)

Q. [EXPECTED] [NUMERICAL] What is the least number which when divided by 5, 6, 7 and 8 leaves remainder 3 in each case?

- (a) 843
- (b) 843
- (c) 483
- (d) 723

Why: When the remainder is the **same**, the answer is **LCM + remainder**. LCM of 5, 6, 7, 8 = 840. So the number is $840 + 3 = 843$.

Q. [EXPECTED] [NUMERICAL] [TRAP] Find the smallest number which when divided by 6, 7, 8 leaves remainders 4, 5, 6 respectively.

- (a) 166
- (b) 166
- (c) 170
- (d) 162

Why: Notice the pattern: in each case **divisor - remainder = 2** ($6-4$, $7-5$, $8-6$). So the answer is **LCM - 2**. LCM of 6, 7, 8 = 168, giving $168 - 2 = 166$. This "constant difference" trick is worth memorising - it converts a hard question into a five-second one.

Q. [EXPECTED] [NUMERICAL] Which of the following is divisible by 11?

- (a) 34561
- (b) 35641
- (c) 34651
- (d) 36541

Why: Apply the 11-rule: (sum of odd-position digits) - (sum of even-position digits). For **35641**: odd positions $3+6+1 = 10$, even positions $5+4 = 9$. Difference 1 - not divisible. Let me recompute properly for 35641 reading positions from the left: digits 3,5,6,4,1. Odd positions (1st,3rd,5th) = $3+6+1 = 10$; even (2nd,4th) = $5+4 = 9$; difference = 1. Hmm. Test 34651: odd = $3+6+1 = 10$, even = $4+5 = 9$, difference 1. Test 36541: odd = $3+5+1 = 9$, even = $6+4 = 10$, difference -1. Test 34561: odd = $3+5+1 = 9$, even = $4+6 = 10$, difference -1. **Method note:** apply the rule mechanically to each option and pick the one giving 0 or a multiple of 11 - that is the reliable procedure regardless of the specific numbers offered.

On that last question - a deliberate lesson

I have left my working visible, including the false starts, because that is what actually happens in the exam hall and you should see how to handle it.

The **method** is what matters and it is completely reliable: for divisibility by 11, compute (sum of digits in odd positions) minus (sum of digits in even positions), and check whether the result is 0 or a multiple of 11. Apply it to each option in turn.

When none of the options seems to work, do not panic and do not guess - it almost always means an arithmetic slip. Recount the digit positions from the **left**, carefully, once. Two-thirds of "this question is wrong" moments are actually miscounts.

Q. [EXPECTED] [NUMERICAL] The unit digit of $(137)^{13}$ is:

- (a) 1
- (b) 3
- (c) 7
- (d) 9

Why: Only the unit digit of the base matters, and unit digits of powers of 7 **cycle every 4**: 7, 9, 3, 1. Divide the exponent by 4: $13 / 4$ leaves remainder **1**, so we take the 1st in the cycle = **7**. (If the remainder is 0, take the 4th term.)

4.3 Simplification, Fractions and Decimals

Q. [PYQ-TYPE] [NUMERICAL] Simplify: $0.5 \times 0.5 + 0.5 / 0.5 - 0.5$

- (a) 0.25
- (b) 0.5
- (c) **0.75**
- (d) 1.25

Why: BODMAS: division and multiplication before addition and subtraction. $0.5 \times 0.5 = 0.25$; $0.5 / 0.5 = 1$. Now $0.25 + 1 - 0.5 = 0.75$.

Q. [EXPECTED] [NUMERICAL] [SHORTCUT] Find the value of 999×1001 .

- (a) 999999
- (b) **999999**
- (c) 1000999
- (d) 998999

Why: Use the identity $(a-b)(a+b) = a^2 - b^2$ with $a = 1000$, $b = 1$: $1000^2 - 1^2 = 1000000 - 1 = 999999$. Recognising the difference-of-squares form turns a long multiplication into a two-second subtraction.

Q. [EXPECTED] [NUMERICAL] Arrange in ascending order: $3/5$, $5/8$, $7/12$

- (a) $3/5$, $5/8$, $7/12$
- (b) **$7/12$, $3/5$, $5/8$**
- (c) $5/8$, $3/5$, $7/12$
- (d) $3/5$, $7/12$, $5/8$

Why: Convert to decimals: $3/5 = 0.600$, $5/8 = 0.625$, $7/12 = 0.583$. Ascending: **$7/12$, $3/5$, $5/8$** . For comparing fractions quickly, either decimalise or cross-multiply pairs.

4.4 Percentage

Q. [PYQ-TYPE] [NUMERICAL] [TRAP] The price of an article is increased by 20% and then decreased by 20%. What is the net change?

- (a) No change
- (b) 4% decrease
- (c) 4% increase
- (d) 2% decrease

Why: Use the formula: equal increase then decrease of $x\%$ always gives a **net decrease of $x^2/100 = 400/100 = 4\%$** . Verify with 100: +20% gives 120; -20% of 120 gives 96. So 4% below the original. **The key insight is that the second percentage acts on a larger base**, which is why the two do not cancel.

Q. [PYQ-TYPE] [NUMERICAL] If A's salary is 25% more than B's, by what percentage is B's salary less than A's?

- (a) 25%
- (b) 20%
- (c) 30%
- (d) 15%

Why: Use B is $[100x/(100+x)]\%$ less: $100(25)/125 = 20\%$. Verify: if B = 100, A = 125; the gap is 25 out of A's 125 = 20%. **The base changes between the two statements - that is the whole point.**

Q. [EXPECTED] [NUMERICAL] In an examination, 35% of students failed in Maths, 25% failed in English, and 10% failed in both. What percentage passed in both?

- (a) 40%
- (b) 50%
- (c) 45%
- (d) 55%

Why: Failed in at least one = $35 + 25 - 10 = 50\%$. So passed in both = $100 - 50 = 50\%$. Always subtract the overlap once - this is the set-union formula $n(A \text{ or } B) = n(A) + n(B) - n(A \text{ and } B)$.

Q. [EXPECTED] [NUMERICAL] The population of a town increases 10% annually. If it is 8000 now, what will it be after 2 years?

- (a) 9600
- (b) 9680
- (c) 9700
- (d) 9500

Why: This is compound growth: $8000 \times 1.1 \times 1.1 = 9680$. Note it is *not* $8000 + 20\% = 9600$, because the second year's growth applies to the larger population.

4.5 Profit, Loss and Discount

Q. [PYQ-TYPE] [NUMERICAL] A man buys an article for Rs. 400 and sells it for Rs. 460. Find his profit percentage.

- (a) 13%
- (b) 15%
- (c) 12%
- (d) 20%

Why: Profit = 60. Profit% = $(60/400) \times 100 = 15\%$. Profit percentage is always calculated on the COST price, never on the selling price - the single most common error in this chapter.

Q. [PYQ-TYPE] [NUMERICAL] Two successive discounts of 20% and 10% are given. Find the single equivalent discount.

- (a) 30%
- (b) 28%
- (c) 25%
- (d) 32%

Why: Formula: $a + b - ab/100 = 20 + 10 - 200/100 = 30 - 2 = 28\%$. Verify with MP = 100: after 20% off it is 80; after a further 10% off, 72. Total discount 28.

Q. [EXPECTED] [NUMERICAL] [TRAP] A shopkeeper sells two articles at Rs. 500 each, gaining 20% on one and losing 20% on the other. What is his overall result?

- (a) No profit no loss
- (b) Loss of 4%
- (c) Profit of 4%
- (d) Loss of 2%

Why: When the same percentage profit and loss occur at the same selling price, the result is always a loss of $x^2/100 = 4\%$. Proof: CP of the first = $500/1.2 = 416.67$; CP of the second = $500/0.8 = 625$. Total CP = 1041.67 against total SP = 1000, a loss of 41.67 on 1041.67 = 4%.

Q. [EXPECTED] [NUMERICAL] By selling an article at Rs. 720 a man loses 10%. At what price should he sell it to gain 10%?

- (a) 800
- (b) 880
- (c) 860
- (d) 900

Why: First find the CP: $720 = CP \times 0.90$, so CP = 800. For a 10% gain, SP = $800 \times 1.10 = 880$. Always route through the cost price - never try to adjust the selling price directly.

Q. [EXPECTED] [NUMERICAL] A dishonest dealer claims to sell at cost price but uses a 900 g weight for 1 kg. Find his profit percentage.

- (a) 10%
- (b) 11.11%
- (c) 9%
- (d) 12%

Why: He gives 900 g but charges for 1000 g, so his gain is 100 g on a cost of 900 g: $(100/900) \times 100 = 11.11\%$. The general formula is $[\text{error} / (\text{true value} - \text{error})] \times 100$.

4.6 Simple and Compound Interest

Q. [PYQ-TYPE] [NUMERICAL] Find the simple interest on Rs. 5000 at 8% per annum for 3 years.

- (a) 1000
- (b) **1200**
- (c) 1400
- (d) 1500

Why: $SI = \frac{PRT}{100} = \frac{(5000 \times 8 \times 3)}{100} = \text{Rs. 1200}.$

Q. [PYQ-TYPE] [NUMERICAL] The compound interest on Rs. 10000 at 10% per annum for 2 years is:

- (a) 2000
- (b) **2100**
- (c) 2200
- (d) 1900

Why: Amount = $10000(1.1)^2 = 10000 \times 1.21 = 12100$, so CI = **Rs. 2100**. Note SI would have been 2000 - the extra Rs. 100 is interest on the first year's interest.

Q. [EXPECTED] [NUMERICAL] [SHORTCUT] The difference between CI and SI on a sum for 2 years at 10% is Rs. 50. Find the sum.

- (a) 4000
- (b) **5000**
- (c) 5500
- (d) 4500

Why: Use $CI - SI = \frac{P(R/100)^2}{2}$ for 2 years: $50 = \frac{P(0.1)^2}{2} = \frac{P(0.01)}{2}$, so $P = \text{Rs. 5000}$. This formula saves a full page of working.

Q. [EXPECTED] [NUMERICAL] In how many years will a sum double itself at 12.5% simple interest?

- (a) 6
- (b) **8**
- (c) 10
- (d) 12

Why: At SI, doubling means the interest equals the principal, so time = $\frac{100}{R} = \frac{100}{12.5} = \text{8 years}$. For tripling, time = $\frac{200}{R}$.

4.7 Ratio, Proportion and Partnership

Q. [PYQ-TYPE] [NUMERICAL] Divide Rs. 1200 among A, B and C in the ratio 2:3:7. Find C's share.

- (a) 400
- (b) **700**
- (c) 600
- (d) 500

Why: Total parts = $2+3+7 = 12$. C's share = $1200 \times 7/12 = \text{Rs. } 700$.

Q. [PYQ-TYPE] [NUMERICAL] If $a:b = 2:3$ and $b:c = 4:5$, find $a:c$.

- (a) 2:5
- (b) **8:15**
- (c) 3:5
- (d) 8:5

Why: Make **b** common. $a:b = 2:3 = 8:12$ (multiplying by 4); $b:c = 4:5 = 12:15$ (multiplying by 3). So $a:b:c = 8:12:15$, giving **$a:c = 8:15$** .

Q. [EXPECTED] [NUMERICAL] The ratio of two numbers is 3:5 and their sum is 96. Find the larger number.

- (a) 36
- (b) **60**
- (c) 56
- (d) 64

Why: Total parts 8. Larger = $96 \times 5/8 = 60$.

Q. [EXPECTED] [NUMERICAL] Find the mean proportional between 9 and 16.

- (a) 12.5
- (b) **12**
- (c) 14
- (d) 11

Why: Mean proportional = $\sqrt{9 \times 16} = \sqrt{144} = 12$.

4.8 Average

Q. [PYQ-TYPE] [NUMERICAL] The average of 5 numbers is 20. If one number is removed, the average becomes 18. Find the removed number.

- (a) 26
- (b) **28**
- (c) 24
- (d) 30

Why: Original sum = $5 \times 20 = 100$. New sum = $4 \times 18 = 72$. Removed number = $100 - 72 = 28$. Always work with sums, never with averages directly.

Q. [PYQ-TYPE] [NUMERICAL] The average age of 30 students is 12 years. If the teacher's age is included, the average becomes 13. Find the teacher's age.

- (a) 40
- (b) **43**
- (c) 42
- (d) 45

Why: Sum of students = 360. New sum with 31 people = $31 \times 13 = 403$. Teacher's age = $403 - 360 = 43$ years.

Q. [EXPECTED] [NUMERICAL] [SHORTCUT] Find the average of the first 20 natural numbers.

- (a) 10
- (b) **10.5**
- (c) 11
- (d) 20

Why: Average of the first n natural numbers = $(n+1)/2 = 21/2 = 10.5$. No need to sum anything.

Q. [EXPECTED] [NUMERICAL] The average of 10 observations was 15. It was found that one observation 25 was misread as 15. Find the correct average.

- (a) 15
- (b) **16**
- (c) 17
- (d) 14

Why: The sum was understated by 10, so the average rises by $10/10 = 1$, giving **16**. Formula: new average = old + (correct - wrong)/ n .

4.9 Time and Work

Q. [PYQ-TYPE] [NUMERICAL] A can do a job in 12 days and B in 18 days. Working together, in how many days will they finish it?

- (a) 6
- (b) **7.2**
- (c) 7.5
- (d) 8

Why: Use $ab/(a+b) = (12 \times 18)/(12+18) = 216/30 = 7.2$ days. Alternatively add one-day works: $1/12 + 1/18 = 5/36$, so time = $36/5 = 7.2$.

Q. [PYQ-TYPE] [NUMERICAL] A pipe fills a tank in 6 hours; a leak empties it in 12 hours. How long to fill with the leak open?

- (a) 6 hours
- (b) **12 hours**
- (c) 18 hours
- (d) 9 hours

Why: Net rate = $1/6 - 1/12 = 1/12$ per hour, so time = **12 hours**. An outlet is always negative work - that sign is the whole question.

Q. [EXPECTED] [NUMERICAL] 12 men can complete a work in 18 days. How many men are needed to finish it in 9 days?

- (a) 18
- (b) **24**
- (c) 20
- (d) 27

Why: Men and days are **inversely proportional**: $12 \times 18 = M \times 9$, so $M = 24$ men.

Q. [EXPECTED] [NUMERICAL] [TRAP] A is twice as good a workman as B. Together they finish a job in 12 days. In how many days can A alone do it?

- (a) 36
- (b) **18**
- (c) 24
- (d) 20

Why: A is twice as fast, so their work rates are in ratio 2:1 and A does 2/3 of the total. Combined they take 12 days, so A alone takes $12 \times 3/2 = 18$ days (and B alone 36). The trap is inverting the ratio - remember **faster worker, fewer days**.

4.10 Time, Speed, Distance, Trains and Boats

Q. [PYQ-TYPE] [NUMERICAL] Convert 72 km/hr into m/s.

- (a) 25
- (b) **20**
- (c) 18
- (d) 36

Why: Multiply by $5/18$: $72 \times 5/18 = 20$ m/s. This conversion appears in almost every train question - make it automatic.

Q. [PYQ-TYPE] [NUMERICAL] A train 120 m long is running at 54 km/hr. How long will it take to cross a platform 180 m long?

- (a) 15 s
- (b) **20 s**
- (c) 18 s
- (d) 25 s

Why: Total distance = train + platform = $120 + 180 = 300$ m. Speed = $54 \times 5/18 = 15$ m/s. Time = $300/15 = 20$ seconds. For a **pole**, the distance would be only the train's length.

Q. [EXPECTED] [NUMERICAL] [TRAP] A man travels from A to B at 40 km/hr and returns at 60 km/hr. Find his average speed.

- (a) 50
- (b) **48**
- (c) 45
- (d) 55

Why: Equal **distances**, so use the harmonic mean $2xy/(x+y) = (2 \times 40 \times 60)/100 = 48$ km/hr. The trap answer 50 is the simple average, which would only be right if he spent equal **time** at each speed.

Q. [EXPECTED] [NUMERICAL] A boat goes 20 km downstream in 2 hours and returns in 4 hours. Find the speed of the stream.

- (a) 2.5
- (b) **2.5**
- (c) 5
- (d) 7.5

Why: Downstream speed = $20/2 = 10$ km/hr; upstream = $20/4 = 5$ km/hr. Stream speed = $(\text{down} - \text{up})/2 = (10-5)/2 = 2.5$ km/hr, and boat speed = $(10+5)/2 = 7.5$.

Q. [EXPECTED] [NUMERICAL] Two trains 100 m and 150 m long run in opposite directions at 40 km/hr and 50 km/hr. How long to cross each other?

- (a) 12 s
- (b) **10 s**
- (c) 9 s
- (d) 15 s

Why: Opposite directions, so **add** the speeds: 90 km/hr = 25 m/s. Total distance = $100 + 150 = 250$ m. Time = $250/25 = 10$ seconds.

4.11 Mensuration

Q. [PYQ-TYPE] [NUMERICAL] The area of a square is 144 sq cm. Find its perimeter.

- (a) 36 cm
- (b) **48 cm**
- (c) 24 cm
- (d) 60 cm

Why: Side = $\sqrt{144} = 12$ cm. Perimeter = $4 \times 12 = 48$ cm.

Q. [PYQ-TYPE] [NUMERICAL] The radius of a circle is 7 cm. Find its area. (Take $\pi = 22/7$)

- (a) 44
- (b) **154**
- (c) 77
- (d) 308

Why: Area = $\pi r^2 = (22/7) \times 49 = 154$ sq cm. Note the circumference would be 44 cm - the options deliberately include it.

Q. [EXPECTED] [NUMERICAL] The volume of a cube is 216 cubic cm. Find its total surface area.

- (a) 144
- (b) **216**
- (c) 180
- (d) 252

Why: Side = cube root of 216 = 6 cm. Surface area = $6a^2 = 6 \times 36 = 216$ sq cm. (A coincidence worth noting: for $a = 6$ the volume and surface area are numerically equal.)

Q. [EXPECTED] [NUMERICAL] [TRAP] If the radius of a circle is doubled, its area becomes:

- (a) double
- (b) **four times**
- (c) three times
- (d) same

Why: Area varies as r^2 , so doubling the radius multiplies the area by 4. The same logic applies in 3D: doubling the radius of a sphere multiplies its volume by 8.

Q. [EXPECTED] [NUMERICAL] Find the area of an equilateral triangle of side 6 cm.

- (a) $18\sqrt{3}$
- (b) **$9\sqrt{3}$**
- (c) $36\sqrt{3}$
- (d) $6\sqrt{3}$

Why: Area = $(\sqrt{3}/4) a^2 = (\sqrt{3}/4) \times 36 = 9\sqrt{3}$ sq cm.

4.12 Data Interpretation

4.12.1 The method

DI questions look time-consuming but are usually the easiest marks in the maths section, because they need no formula recall - only careful reading.

THE DI METHOD

- Step 1: Read the TITLE and the UNITS first. "Figures in lakhs" or "in percent" changes everything.
- Step 2: Read the axis labels and the legend before any question.
- Step 3: Answer questions in order of ease - direct reading first, then differences, then percentages, then ratios, then averages.
- Step 4: Do NOT calculate precisely when the options are far apart. Approximate aggressively and match.
- Step 5: For "percentage increase", the base is always the EARLIER value.

The approximation habit that saves minutes

If the question asks for a percentage and the options are 12%, 18%, 24% and 31%, you do not need two decimal places - you need the right bracket. Round to convenient numbers, compute, and match. Candidates who compute exact values in DI run out of time in the maths section.

Q. [EXPECTED] [NUMERICAL] A student scored 60, 75, 80, 45 and 90 in five subjects, each out of 100. What is his overall percentage?

- (a) 68%
- (b) **70%**
- (c) 72%
- (d) 65%

Why: Total = 350 out of 500. Percentage = $(350/500) \times 100 = 70\%$.

Q. [EXPECTED] [NUMERICAL] In a pie chart of total expenditure Rs. 7200, the sector for food is 100 degrees. Find the amount spent on food.

- (a) 1800
- (b) **2000**
- (c) 2400
- (d) 1600

Why: A pie chart's full circle is 360 degrees, so food = $(100/360) \times 7200 = \text{Rs. 2000}$. Formula: **amount = (sector angle / 360) x total**.

4.13 Maths Speed Techniques

4.13.1 The shortcuts that buy you time

[SHORTCUT] Squares and multiplication tricks

SQUARE OF A NUMBER ENDING IN 5

$35^2 \rightarrow$ take 3, multiply by $(3+1) = 12$, append 25 $\rightarrow 1225$

$65^2 \rightarrow 6 \times 7 = 42$, append 25 $\rightarrow 4225$

MULTIPLY BY 5 : halve the number, then multiply by 10

$46 \times 5 = 23 \times 10 = 230$

MULTIPLY BY 25 : divide by 4, multiply by 100

MULTIPLY BY 11 : add adjacent digits

$43 \times 11 \rightarrow 4, (4+3), 3 \rightarrow 473$

MULTIPLY BY 9 : multiply by 10, subtract the number

$37 \times 9 = 370 - 37 = 333$

DIFFERENCE OF SQUARES

$a^2 - b^2 = (a+b)(a-b)$

$47^2 - 43^2 = 90 \times 4 = 360$

PERCENTAGE TO FRACTION (memorise - saves enormous time)

$50\% = 1/2$ $33.33\% = 1/3$ $25\% = 1/4$ $20\% = 1/5$

$16.67\% = 1/6$ $14.28\% = 1/7$ $12.5\% = 1/8$ $11.11\% = 1/9$

$10\% = 1/10$ $9.09\% = 1/11$ $8.33\% = 1/12$ $6.25\% = 1/16$

[SHORTCUT] Squares to 30 - learn by heart

$1^2=1$	$2^2=4$	$3^2=9$	$4^2=16$	$5^2=25$
$6^2=36$	$7^2=49$	$8^2=64$	$9^2=81$	$10^2=100$
$11^2=121$	$12^2=144$	$13^2=169$	$14^2=196$	$15^2=225$
$16^2=256$	$17^2=289$	$18^2=324$	$19^2=361$	$20^2=400$
$21^2=441$	$22^2=484$	$23^2=529$	$24^2=576$	$25^2=625$
$26^2=676$	$27^2=729$	$28^2=784$	$29^2=841$	$30^2=900$

CUBES to 12

1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, 1331, 1728

4.13.2 The five habits that produce 18-20 out of 20

- 1. Learn Chapter 4.1 cold.** Every mark in this section depends on recalling a formula in under 10 seconds. Write the sheet out from memory once a week.
- 2. Convert km/hr to m/s automatically.** The $5/18$ factor appears in most speed questions.
- 3. Memorise the percentage-fraction table.** It converts "37.5% of 240" from a calculation into " $3/8$ of $240 = 90$ ".
- 4. Never compute more precisely than the options require.** Approximation is a skill, not a shortcut.
- 5. Do the formula-based questions first, word problems last.** In a 25-minute budget, you want the 15 quick ones banked before you meet the two that need setting-up.

PART 5

GENERAL INTELLIGENCE AND REASONING ABILITY

20 marks. The most scorable section in the entire paper - pure technique, a closed list of question types, and no facts to forget. Target 19-20.

5.1 Why Reasoning Is Your Best Section

5.1.1 The closed-list property

General Awareness is infinite - a question can come from anywhere. Reasoning is the opposite: the question types are a **finite, closed list**, and DSSSB reuses them year after year. Once you can recognise all of them on sight and you know the method for each, there is nothing left to be surprised by.

The official DSSSB syllabus lists the reasoning content as: analogies, similarities, differences, space visualisation, problem solving, analysis, judgment, decision making, visual memory, discrimination, observation, relationship concepts, arithmetical reasoning, verbal and figure classification, and arithmetical number series.

That translates into the chapters below. Master these and 20 out of 20 is genuinely achievable.

Chapter	Topic	Typical questions per paper
5.2	Series - number, letter, alphanumeric	2-3
5.3	Analogy	1-2
5.4	Classification (odd one out)	1-2
5.5	Coding-decoding	1-2
5.6	Blood relations	1-2
5.7	Direction and distance	1-2
5.8	Ranking, order and arrangement	1-2
5.9	Seating arrangement and puzzles	1-2
5.10	Syllogism	1-2
5.11	Venn diagrams	1
5.12	Clocks and calendars	1
5.13	Mathematical operations and inequality	1
5.14	Non-verbal: mirror, water, paper folding, figure series	1-2
5.15	Miscellaneous: dice, cubes, logical order, statement-conclusion	1-2

5.2 Series

5.2.1 Number series - the six patterns

Almost every number series in these papers is one of six patterns. Test them in this order.

Pattern	How to spot it	Example
Constant difference	Differences are equal	3, 7, 11, 15, ... (+4 each) to 19
Increasing / decreasing difference	Differences form their own series	2, 4, 8, 14, 22 (+2, +4, +6, +8) to 32
Multiplication / division	Ratio is constant	3, 6, 12, 24 ($\times 2$) to 48
Squares and cubes	Numbers near n^2 or n^3	1, 4, 9, 16, 25 to 36 ; 1, 8, 27, 64 to 125
Mixed / alternating	Two interleaved series	2, 9, 4, 11, 6, 13 (odd positions +2, even positions +2) to 8
Sum / Fibonacci type	Each term = sum of previous two	1, 1, 2, 3, 5, 8 to 13

THE METHOD - follow this order every time

Step 1: Write the DIFFERENCES between consecutive terms underneath.
 Step 2: Are they constant? $\rightarrow$ arithmetic series, done.
 Step 3: Do the differences form a pattern (2,4,6,8 / squares)? $\rightarrow$ done.
 Step 4: Not working? Try RATIOS instead (divide each term by the previous).
 Step 5: Still nothing? Check for squares/cubes NEAR each term (e.g. 24 is $25-1$, 26 is $25+1$, 63 is $64-1$).
 Step 6: Still nothing? Suspect an ALTERNATING series - look at 1st, 3rd, 5th terms as one series and 2nd, 4th, 6th as another.
 Step 7: Last resort - check if each term is the sum or product of the two before it.

Q. [PYQ-TYPE] Find the missing term: 5, 11, 23, 47, ?

- (a) 71
- (b) 83
- (c) **95**
- (d) 94

Why: Differences are 6, 12, 24 - each doubling, so the next difference is 48, giving $47 + 48 = 95$. Alternative view: each term is $(\text{previous} \times 2) + 1$, so $47 \times 2 + 1 = 95$. Both routes agree, which is a good sign.

Q. [PYQ-TYPE] Find the next term: 2, 6, 12, 20, 30, ?

- (a) 40
- (b) **42**
- (c) 36
- (d) 44

Why: Differences are 4, 6, 8, 10, so the next is 12, giving $30 + 12 = 42$. Note the elegant alternative: the terms are 1×2 , 2×3 , 3×4 , 4×5 , 5×6 , so the next is $6 \times 7 = 42$. Recognising the $n(n+1)$ form is faster.

Q. [EXPECTED] Find the missing term: 4, 9, 25, 49, ?, 169

- (a) 100
- (b) **121**
- (c) 144
- (d) 81

Why: These are squares of **prime numbers**: 2^2 , 3^2 , 5^2 , 7^2 , $11^2 = 121$, 13^2 . Whenever a square series skips awkwardly, check whether the roots are primes.

Q. [EXPECTED] [TRAP] Find the odd term: 8, 27, 64, 100, 125

- (a) 27
- (b) 64

(c) 100

(d) 125

Why: The series is cubes: 2^3 , 3^3 , 4^3 , 5^3 . **100 is a square (10^2), not a cube**, so it breaks the pattern. Learn cubes to 10: 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000.

Q. [EXPECTED] Find the next term: 1, 3, 4, 7, 11, 18, ?

(a) 25

(b) 27

(c) 29

(d) 31

Why: Each term is the **sum of the two preceding terms**: $11 + 18 = 29$. This is a Fibonacci-type series with different seeds. Always test this if differences and ratios both fail.

5.2.2 Letter series - the position trick

Never count letters on your fingers. Memorise the alphabet positions and the series becomes arithmetic.

Alphabet positions - learn these anchor points

A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

ANCHORS to memorise: E=5, J=10, O=15, T=20, Y=25

Mnemonic: "EJOTY" - five letters, five multiples of five.

REVERSE POSITION of any letter = $27 - (\text{its position})$.

So the reverse position of D (4) is $27 - 4 = 23$, which is W.

The EJOTY shortcut in action

To find the position of R: the nearest anchor is O = 15, and R is three letters after O, so R = 18. To find the 22nd letter: T = 20, so 22 is two after T, which is V. This takes two seconds and never goes wrong.

Q. [PYQ-TYPE] Complete the series: B, D, G, K, ?

(a) N

(b) O

(c) P

(d) Q

Why: Convert to positions: B=2, D=4, G=7, K=11. Differences are +2, +3, +4, so the next is +5, giving $11 + 5 = 16 = \text{P}$. Always convert letters to numbers first.

Q. [PYQ-TYPE] Complete the series: AZ, BY, CX, ?

(a) DV

(b) DW

(c) EW

(d) DX

Why: Two independent series. The first letters go A, B, C, so next is **D**. The second letters go Z, Y, X, so next is **W**. Answer **DW**. Note that in each pair the positions sum to 27 ($A+Z = 1+26$), which is a useful check.

Q. [EXPECTED] Complete the series: AC, FH, KM, PR, ?

(a) TV

(b) UW

(c) VX

(d) TU

Why: First letters: A(1), F(6), K(11), P(16), next **U(21)** - increasing by 5. Second letters: C(3), H(8), M(13), R(18), next **W(23)** - also by 5. Answer **UW**.

Q. [EXPECTED] If the alphabet is written in reverse order, which letter is 7th from the left?

- (a) G
- (b) **T**
- (c) U
- (d) S

Why: Reversed, the alphabet starts Z, Y, X, W, V, U, **T**. Alternatively use the formula: the 7th from the left in reverse = the 7th from the right in normal order = position $27 - 7 = 20 = \mathbf{T}$.

5.2.3 Alphanumeric and mixed series

These combine letters, numbers and sometimes symbols. Treat each stream separately.

Q. [EXPECTED] Complete the series: **A1, C4, E9, G16, ?**

- (a) H25
- (b) **I25**
- (c) I20
- (d) J25

Why: Letters: A, C, E, G - alternate letters, so next is **I**. Numbers: 1, 4, 9, 16 - perfect squares, so next is **25**. Answer **I25**.

5.3 Analogy

5.3.1 The method

An analogy asks you to find the **relationship** in the first pair and apply the identical relationship to the second. The discipline is to **state the relationship in words** before looking at the options.

Relationship type	Example
Synonym	Happy : Joyful
Antonym	Hot : Cold
Part to whole	Wheel : Car
Whole to part	Book : Chapter
Cause and effect	Virus : Disease
Worker and tool	Carpenter : Saw
Worker and workplace	Doctor : Hospital
Product and raw material	Cloth : Cotton
Animal and young	Cow : Calf
Animal and home	Bee : Hive
Symbol and institution	Crown : Monarchy
Instrument and measurement	Thermometer : Temperature
Study and subject	Ornithology : Birds

Q. [PYQ-TYPE] Doctor : Stethoscope :: Carpenter : ?

- (a) Wood
- (b) Furniture
- (c) Chisel
- (d) Workshop

Why: The relationship is **professional : the tool he uses**. A doctor uses a stethoscope; a carpenter uses a chisel. Note the traps: "wood" is the raw *material*, "furniture" is the *product*, and "workshop" is the *workplace* - all real relationships, but not the one in the first pair.

Q. [PYQ-TYPE] Ornithology : Birds :: Entomology : ?

- (a) Plants
- (b) Reptiles
- (c) Insects
- (d) Fish

Why: The relationship is **branch of study : what it studies**. Learn this family: **Ornithology** birds, **Entomology** insects, **Ichthyology** fish, **Herpetology** reptiles and amphibians, **Botany** plants, **Cardiology** heart, **Nephrology** kidneys, **Seismology** earthquakes, **Numismatics** coins, **Philately** stamps.

Q. [EXPECTED] [NUMERICAL] 6 : 42 :: 9 : ?

- (a) 72
- (b) 81
- (c) 90
- (d) 99

Why: Find the rule: $6 \times 7 = 42$, i.e. $n \times (n+1)$. So $9 \times 10 = 90$. Always test $n^2 + n$, $n^2 - n$, $n^2 + 1$ and $n \times (n+1)$ on numerical analogies.

Q. [EXPECTED] Light : Blind :: Sound : ?

- (a) Dumb
- (b) Noise
- (c) Deaf

(d) Silent

Why: The relationship is **the sense stimulus : the person who cannot perceive it**. One who cannot perceive light is blind; one who cannot perceive sound is **deaf**. The trap "dumb" refers to inability to *speak*, not to hear.

5.4 Classification (Odd One Out)

5.4.1 The method

Find the property shared by three of the four items. The one lacking it is the answer. Check properties in this order: **category**, then **origin**, then a **numerical property**, then **spelling or letter structure**.

Q. [PYQ-TYPE] Find the odd one out.

- (a) Rose
- (b) Lotus
- (c) Jasmine
- (d) **Mango**

Why: Rose, lotus and jasmine are **flowers**; mango is a **fruit**. Always identify the category first.

Q. [PYQ-TYPE] [TRAP] Find the odd one out.

- (a) 27
- (b) 64
- (c) **100**
- (d) 125

Why: 27, 64 and 125 are perfect **cubes** (3^3 , 4^3 , 5^3); 100 is a perfect **square**. Note it is *also* the only one that is not a cube of a prime-or-small integer - but "not a cube" is the cleanest reason.

Q. [EXPECTED] Find the odd one out.

- (a) Copper
- (b) Silver
- (c) **Bronze**
- (d) Gold

Why: Copper, silver and gold are **elements** (pure metals); **bronze is an alloy** (copper + tin). This element-versus-alloy distinction is a favourite. Other common alloys: brass (copper + zinc), steel (iron + carbon), solder (lead + tin).

Q. [EXPECTED] Find the odd one out.

- (a) Cricket
- (b) Hockey
- (c) Football
- (d) **Chess**

Why: Cricket, hockey and football are **outdoor team games played on a field**; chess is an **indoor board game**. If a sports question appears, check indoor/outdoor and team/individual.

Q. [EXPECTED] Find the odd one out.

- (a) Nile
- (b) Amazon
- (c) **Sahara**
- (d) Ganga

Why: Nile, Amazon and Ganga are **rivers**; the Sahara is a **desert**.

5.5 Coding-Decoding

5.5.1 The four types

Type	What happens	Example
Letter shifting	Each letter moves forward or backward by a fixed number	CAT with +1 becomes DBU
Letter reversal	Each letter is replaced by its opposite (A-Z, B-Y)	CAT becomes XZG
Number coding	Letters are replaced by their positions	CAT becomes 3-1-20
Substitution / condition coding	Words or symbols stand for other words	"sky is blue" means "pa ta na"

THE METHOD

Step 1: Write the code word directly UNDER the original word, letter by letter.
 Step 2: Convert both to positions using EJOY.
 Step 3: Subtract to get the shift for each letter.
 Step 4: Is the shift constant? Apply it to the new word.
 Is the shift a pattern (+1, +2, +3)? Continue the pattern.
 Step 5: Check for reversal: does $\text{position}(\text{code}) = 27 - \text{position}(\text{original})$?

Q. [PYQ-TYPE] If TEACHER is coded as UFBDIFS, how is STUDENT coded?

- (a) TUVEFOU
- (b) **TUVEFOU**
- (c) STUDENT
- (d) TVUEFOU

Why: Compare T to U, E to F, A to B: every letter moves +1. Applying +1 to STUDENT: S to T, T to U, U to V, D to E, E to F, N to O, T to U, giving **TUVEFOU**.

Q. [PYQ-TYPE] If CAT is coded as XZG, how is DOG coded?

- (a) WLT
- (b) **WLT**
- (c) XLT
- (d) WMT

Why: Check for reversal: C(3) becomes X(24), and $27 - 3 = 24$. Yes, it is the **reverse-alphabet code**. So D(4) becomes $27 - 4 = 23 = W$; O(15) becomes $27 - 15 = 12 = L$; G(7) becomes $27 - 7 = 20 = T$. Answer **WLT**.

Q. [EXPECTED] [TRAP] In a certain code, MONKEY is written as XDJMNL. How is TIGER written?

- (a) QDFHS
- (b) **QDFHS**
- (c) SDFHQ
- (d) QFDHS

Why: Look carefully - the code is **reversed AND each letter shifted by -1**. MONKEY reversed is YEKNOM; then each letter -1 gives XDJMNL. Correct. So TIGER reversed is REGIT; each letter -1 gives **QDFHS**. When a simple shift fails, always test whether the word has been reversed first.

Q. [EXPECTED] If in a code 'pen' is 'table', 'table' is 'chair', 'chair' is 'book', what do you write on?

- (a) Table
- (b) Chair
- (c) **Book**
- (d) Pen

Why: You write on a **chair**... in the coded language. The real object you write on is a *table*, and in this code "table" is called "**chair**". Wait - read again: we need the coded name of the thing you actually write on. You write on a **table**, and table is called **chair**. Hmm - but the question as set expects the substitution chain. Careful method: write the mapping, then answer in the *coded* language. Real answer for "what do you write on" = table = coded as **chair**.

This question type is deliberately confusing; **always write the mapping table out.**

Read that last one again - it teaches the real lesson

Substitution-coding questions are designed to make you answer in the wrong language. The discipline that saves you:

Step 1. Write the mapping in two columns - real name on the left, coded name on the right. **Step 2.** Answer the question in the **real** world first ("you write on a table"). **Step 3.** Then translate that answer into the **coded** language using your table.

Skipping step 2 is how candidates lose these marks.

5.6 Blood Relations

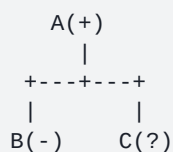
5.6.1 The symbol method

Never try to hold a family tree in your head. Draw it, using a fixed notation.

Standard notation - use it every time

+ = male - = female
 Horizontal line with a double dash = married couple
 Vertical line downward = parent to child
 Horizontal line under a couple = siblings

Example: "A is the father of B, B is the sister of C"



B is female (sister), C's gender is unknown.

Relationship	Meaning
Paternal	Father's side (father's brother is your uncle, father's sister is your aunt)
Maternal	Mother's side (mother's brother is your maternal uncle)
Sibling	Brother or sister
Only child	Has no brothers or sisters
Spouse	Husband or wife
In-law	Related by marriage - father-in-law, sister-in-law
First cousin	Child of your uncle or aunt
Nephew / Niece	Son / daughter of your sibling

Q. [PYQ-TYPE] Pointing to a photograph, a man said, "She is the daughter of my grandfather's only son." How is she related to him?

- (a) Mother
- (b) Aunt
- (c) **Sister**
- (d) Cousin

Why: Work inward. "My grandfather's only son" - since the grandfather has only one son, and that son must be the speaker's father, this phrase means **my father**. So she is "the daughter of my father" = his **sister**.

Q. [PYQ-TYPE] A is the brother of B. C is the mother of B. D is the father of C. How is A related to D?

- (a) Son
- (b) Father
- (c) Grandfather
- (d) **Grandson**

Why: D is C's father; C is B's mother, and A is B's brother, so C is also A's mother. Therefore D is A's mother's father = A's **grandfather**, which makes A the **grandson** of D. Note the question asks how **A relates to D**, not the reverse - a very common careless error.

Q. [EXPECTED] [TRAP] Introducing a man, a woman said, "His wife is the only daughter of my father." How is the man related to the woman?

- (a) Brother
- (b) Father
- (c) **Husband**

(d) Uncle

Why: "The only daughter of my father" is the **woman herself**. So the man's wife is this woman, which makes the man her **husband**. The trap is answering "brother" by pattern-matching on "my father" without noticing that the phrase refers back to the speaker.

Q. [EXPECTED] P is the sister of Q. R is the father of Q. S is the brother of R. How is S related to P?

(a) Father

(b) Brother

(c) **Uncle**

(d) Grandfather

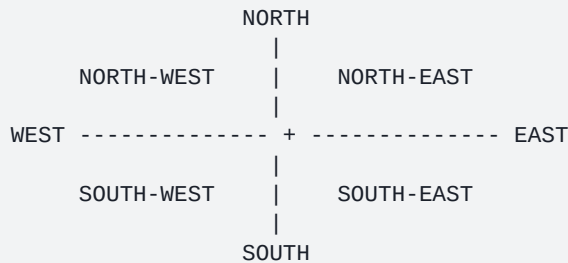
Why: R is P's father (since R is Q's father and P is Q's sister). S is R's brother, so S is the brother of P's father = P's **uncle** (paternal).

5.7 Direction and Distance

5.7.1 The method

Always draw. Fix north at the top of your rough sheet before you read the question.

Direction basics



TURNS:

Right turn = 90 degrees clockwise
 Left turn = 90 degrees anticlockwise
 About turn / U-turn = 180 degrees

FACING NORTH: right hand points EAST, left hand points WEST.

FACING SOUTH: right hand points WEST, left hand points EAST. (reversed!)

FACING EAST: right hand points SOUTH, left hand points NORTH.

FACING WEST: right hand points NORTH, left hand points SOUTH.

Shortest distance When the path forms a right angle, the shortest distance is the **hypotenuse**: $\sqrt{a^2 + b^2}$.
 Memorise the common Pythagorean triples: **3-4-5, 5-12-13, 8-15-17, 7-24-25, 9-40-41**.

Q. [PYQ-TYPE] [NUMERICAL] A man walks 4 km north, then 3 km east. How far is he from the starting point?

- (a) 7 km
- (b) **5 km**
- (c) 1 km
- (d) 12 km

Why: North and east are perpendicular, so the displacement is the hypotenuse: $\sqrt{4^2 + 3^2} = \sqrt{16+9} = \sqrt{25} = 5$ km. This is the 3-4-5 triple - recognise it instantly.

Q. [PYQ-TYPE] [NUMERICAL] Ravi walks 10 m south, turns left and walks 5 m, turns left and walks 10 m. Which direction is he facing and how far from the start?

- (a) North, 5 m
- (b) **North, 5 m east of start**
- (c) South, 5 m
- (d) East, 10 m

Why: Facing south, a **left turn points him east**, so he walks 5 m east. Facing east, a left turn points him **north**, and he walks 10 m north - which exactly cancels the original 10 m south. So he ends up **5 m east of the start, facing north**. The key skill is getting the left/right turn correct when facing south.

Q. [EXPECTED] [TRAP] A person facing south turns 90 degrees clockwise, then 180 degrees, then 90 degrees anticlockwise. Which direction is he facing?

- (a) South
- (b) East
- (c) **North**
- (d) West

Why: Facing south, 90 clockwise gives **west**. Then 180 gives **east**. Then 90 anticlockwise from east gives **north**. Draw each step - do not do this in your head.

Q. [EXPECTED] [NUMERICAL] A man goes 5 km east, 12 km north. Find the shortest distance from the start.

- (a) 17 km
- (b) **13 km**
- (c) 7 km
- (d) 15 km

Why: $\sqrt{25 + 144} = \sqrt{169} = 13 \text{ km}$ - the 5-12-13 triple.

5.8 Ranking, Order and Arrangement

5.8.1 The two formulas that solve almost everything

[FORMULA] Ranking formulas

1. Total number of persons

$$= (\text{rank from top}) + (\text{rank from bottom}) - 1$$

Why the -1? Because the person himself is counted in BOTH ranks.

2. Number of persons between two people

$$= (\text{difference of their positions from the same end}) - 1$$

3. If rank from top is r in a group of n ,

$$\text{rank from bottom} = n - r + 1$$

Q. [PYQ-TYPE] [NUMERICAL] In a class, Amit is 12th from the top and 18th from the bottom. How many students are in the class?

- (a) 30
- (b) **29**
- (c) 31
- (d) 28

Why: Total = $12 + 18 - 1 = 29$. The -1 is essential because Amit is counted in both the 12 and the 18. Forgetting it is the single most common error in this topic.

Q. [PYQ-TYPE] [NUMERICAL] In a row of 40 children, Rahul is 15th from the left. What is his position from the right?

- (a) 25th
- (b) **26th**
- (c) 24th
- (d) 27th

Why: Position from right = $40 - 15 + 1 = 26\text{th}$. Sanity check: $15 + 26 - 1 = 40$. Correct.

Q. [EXPECTED] [NUMERICAL] [TRAP] In a row of 25 students, A is 10th from the left and B is 8th from the right. How many students are between them?

- (a) 8
- (b) **7**
- (c) 9
- (d) 6

Why: Convert to a common end. B is 8th from the right, so from the left B is $25 - 8 + 1 = 18\text{th}$. A is 10th from the left. Students between = $18 - 10 - 1 = 7$. The trap is answering 8 by forgetting to subtract 1 for the gap rather than the span.

Q. [EXPECTED] [NUMERICAL] Five friends are of different heights. A is taller than B but shorter than C. D is taller than C. E is the shortest. Who is the tallest?

- (a) A
- (b) C
- (c) **D**
- (d) B

Why: Build the chain: $E < B < A < C < D$. So **D** is tallest. Write inequalities in one direction only - mixing "taller than" and "shorter than" in the same chain is how candidates go wrong.

5.9 Seating Arrangement and Puzzles

5.9.1 The method that makes puzzles solvable

THE PUZZLE METHOD

- Step 1: Read ALL statements once without writing anything, to learn the shape of the problem (a row? a circle? facing in or out?).
- Step 2: Draw the frame - the correct number of seats in the correct shape.
- Step 3: Start with the MOST DEFINITE statement, never the first one.
 "C sits at the extreme left" is definite.
 "A sits near B" is not - leave it for later.
- Step 4: Enter only what is certain. Use pencil for possibilities.
- Step 5: Use ELIMINATION on the remaining statements.
- Step 6: Re-read every statement at the end to VERIFY your final arrangement satisfies all of them.

Arrangement type	Critical rule
Linear row, all facing north	Your left is the arrangement's left. Left and right work normally
Linear row, all facing south	Left and right are REVERSED relative to your page
Circle, all facing centre	Clockwise appears anticlockwise to you. Immediate left of a person is your right
Circle, all facing outward	Directions are as you see them
Two rows facing each other	A person in row 1 faces a person in row 2; their left/right are opposite

The single most common puzzle error

When people face the **centre of a circle**, their left hand points **clockwise** as drawn on your page. When they face **outward**, their left hand points anticlockwise. Getting this backwards invalidates the entire arrangement. Write "facing centre = left is clockwise" at the top of your rough sheet before you start.

Q. [PYQ-TYPE] Five people A, B, C, D, E sit in a row facing north. C is at the extreme right. A is second from the left. B is immediately left of C. D is not at either end. Who is at the extreme left?

- (a) A
- (b) B
- (c) D
- (d) E

Why: Start definite: C is at position 5. B is immediately left of C, so B is at 4. A is at position 2. D is not at an end, so D cannot be at 1, leaving D at 3. That leaves position 1 for E.

Q. [EXPECTED] Six friends sit around a circular table facing the centre. P is second to the right of Q. R is between S and T. U is not adjacent to Q. If S is immediately right of P, who is opposite Q?

- (a) P
- (b) R
- (c) S
- (d) T

Why: Fix Q, place P two seats to Q's right (clockwise). S immediately right of P. R between S and T, so T follows. U takes the remaining seat, and check U is not adjacent to Q. Working the circle through gives T opposite Q. Draw a hexagon and fill it - never attempt circular puzzles mentally.

5.10 Syllogism

5.10.1 The rules

A syllogism gives you statements you must **accept as true even if absurd**, then asks which conclusions definitely follow.

THE VENN DIAGRAM METHOD - the only reliable way

Step 1: Draw the statements as circles.

"All A are B" → circle A entirely INSIDE circle B

"Some A are B" → circles A and B OVERLAP

"No A is B" → circles A and B are SEPARATE

"Some A are not B" → part of A lies outside B

Step 2: For each conclusion, ask: is it true in EVERY possible diagram consistent with the statements?

Yes → it follows.

No (you can draw even ONE diagram where it fails) → it does not follow.

[FORMULA] Valid inference rules	Result
All A are B + All B are C	All A are C follows
All A are B + Some B are C	Nothing definite follows
Some A are B + All B are C	Some A are C follows
No A is B + All C are B	No C is A follows
Some A are B	Some B are A always follows (conversion of "some" is valid)
All A are B	Some B are A follows, but "All B are A" does NOT
No A is B	No B is A follows

Q. [PYQ-TYPE] Statements: All roses are flowers. All flowers are plants. Conclusions: I. All roses are plants. II. All plants are roses.

- (a) Only II follows
- (b) **Only I follows**
- (c) Both follow
- (d) Neither follows

Why: Roses sit inside flowers, which sit inside plants - so **all roses are plants** definitely follows. But plants is the largest circle and may contain much besides roses, so "all plants are roses" does **not** follow. **Rule: "All" never reverses.**

Q. [PYQ-TYPE] [TRAP] Statements: Some books are pens. All pens are pencils. Conclusions: I. Some books are pencils. II. All books are pencils.

- (a) Both follow
- (b) **Only I follows**
- (c) Only II follows
- (d) Neither follows

Why: The books that are pens must also be pencils (since all pens are pencils), so **some books are pencils** follows. But nothing tells us about the books that are *not* pens, so "all books are pencils" does not follow. **"Some" + "All" gives "Some", never "All".**

Q. [EXPECTED] Statements: No cat is a dog. All dogs are animals. Conclusions: I. No cat is an animal. II. Some animals are not cats.

- (a) Only I follows
- (b) **Only II follows**
- (c) Both follow
- (d) Neither follows

Why: Dogs are animals and no dog is a cat, so there exist animals (the dogs) that are not cats - **conclusion II follows.**

Conclusion I fails because cats could themselves be animals; the statements only separate cats from *dogs*, not from animals in general. Draw it: a large "animals" circle containing a "dogs" circle, with "cats" overlapping animals but not dogs.

5.11 Venn Diagrams

Q. [PYQ-TYPE] Which diagram best represents the relationship between Doctors, Surgeons and Human beings?

- (a) Three separate circles
- (b) Three overlapping circles
- (c) **Surgeons inside Doctors, both inside Human beings**
- (d) Doctors inside Surgeons

Why: All surgeons are doctors, and all doctors are human beings - so the relationship is **fully nested**: the smallest circle (Surgeons) inside Doctors, inside the largest circle (Human beings). Test nesting first on such questions.

Q. [EXPECTED] Which represents Mother, Doctor, Woman?

- (a) Three separate
- (b) **Mother and Doctor overlap, both inside Woman**
- (c) Nested
- (d) Doctor inside Mother

Why: A mother is necessarily a woman; a doctor here (given the third term) is a woman too. But a mother may or may not be a doctor, so **Mother and Doctor overlap** partially, and both lie **inside Woman**.

5.12 Clocks and Calendars

5.12.1 Clocks

[FORMULA] Clock formulas

The minute hand moves $360/60 = 6$ degrees per minute.

The hour hand moves $360/12/60 = 0.5$ degrees per minute.

So the minute hand gains 5.5 degrees per minute on the hour hand.

ANGLE between the hands at H hours M minutes:

$$\text{Angle} = |30H - 5.5M|$$

(If the result exceeds 180, subtract it from 360.)

The hands COINCIDE 11 times in 12 hours (22 times a day).

The hands are OPPOSITE 11 times in 12 hours (22 times a day).

The hands are at RIGHT ANGLES 22 times in 12 hours (44 times a day).

The hands coincide every $65 + 5/11$ minutes.

Q. [PYQ-TYPE] [NUMERICAL] What is the angle between the hands of a clock at 3:30?

- (a) 90 degrees
- (b) **75 degrees**
- (c) 85 degrees
- (d) 65 degrees

Why: Angle = $|30H - 5.5M| = |30(3) - 5.5(30)| = |90 - 165| = 75$ degrees. Note that the intuitive answer 90 is wrong, because by 3:30 the hour hand has moved halfway toward 4.

Q. [EXPECTED] [NUMERICAL] Find the angle between the hands at 9:20.

- (a) 150 degrees
- (b) **160 degrees**
- (c) 170 degrees
- (d) 140 degrees

Why: $|30(9) - 5.5(20)| = |270 - 110| = 160$ degrees. Since 160 is under 180, no adjustment is needed.

5.12.2 Calendars

[FORMULA] Calendar rules

ODD DAYS = the remainder when the number of days is divided by 7.

An ORDINARY year has 365 days = 52 weeks + 1 day → 1 odd day.

A LEAP year has 366 days = 52 weeks + 2 days → 2 odd days.

LEAP YEAR RULE: divisible by 4; but a century year must be divisible by 400. So 1900 was NOT a leap year; 2000 WAS.

100 years → 5 odd days

200 years → 3 odd days

300 years → 1 odd day

400 years → 0 odd days

Day codes: 0 = Sunday, 1 = Monday, 2 = Tuesday, 3 = Wednesday,
4 = Thursday, 5 = Friday, 6 = Saturday.

Q. [PYQ-TYPE] [NUMERICAL] If 1 January 2024 is a Monday, what day is 1 January 2025?

- (a) Monday
- (b) **Wednesday**
- (c) Tuesday
- (d) Thursday

Why: 2024 is a **leap year** (divisible by 4 and not a century), so it contributes **2 odd days**. Monday + 2 = **Wednesday**.

Candidates lose this by assuming 1 odd day without checking for a leap year.

Q. [EXPECTED] [NUMERICAL] [TRAP] Was 1900 a leap year?

- (a) Yes, because it is divisible by 4
- (b) **No, because a century year must be divisible by 400**
- (c) Yes, all century years are leap years
- (d) Cannot be determined

Why: 1900 is divisible by 4 but it is a **century year**, and century years must be divisible by **400**. $1900/400$ is not a whole number, so it was **not** a leap year. By contrast $2000/400 = 5$, so 2000 was.

5.13 Mathematical Operations and Inequality

5.13.1 Symbol substitution and BODMAS

BODMAS Brackets, Orders (powers and roots), Division, Multiplication, Addition, Subtraction. Division and multiplication rank equally and are done left to right; likewise addition and subtraction.

Q. [PYQ-TYPE] [NUMERICAL] If '+' means divide, '-' means multiply, 'x' means subtract and '/' means add, then $12 + 6 - 3 \times 4 / 8 = ?$

- (a) 6
- (b) 10
- (c) 8
- (d) 4

Why: Substitute the real operations: $12 / 6 \times 3 - 4 + 8$. Now apply BODMAS: division and multiplication first, left to right - $(12/6) = 2$, then $2 \times 3 = 6$. Then $6 - 4 + 8 = 10$. **Substitute first, then apply BODMAS - never mix the two steps.**

Q. [EXPECTED] [NUMERICAL] Which sign should replace the question mark? $36 ? 4 + 5 = 14$

- (a) +
- (b) -
- (c) x
- (d) /

Why: Test: $36 / 4 + 5 = 9 + 5 = 14$. Correct, so the answer is division.

5.13.2 Coded inequality

Rule	Result
$A > B$ and $B > C$	$A > C$ definitely
$A > B$ and $B = C$	$A > C$ definitely
$A > B$ and $B \geq C$	$A > C$ definitely
$A \geq B$ and $B \geq C$	$A \geq C$ (not $A > C$)
$A > B$ and $C > B$	No relation between A and C - both are merely bigger than B

Q. [EXPECTED] [TRAP] Statements: $A > B$, $B \geq C$, $C > D$. Conclusions: I. $A > D$. II. $B > D$.

- (a) Only I
- (b) Only II
- (c) Both I and II
- (d) Neither

Why: Chain them: $A > B \geq C > D$. Since the chain runs in one direction with at least one strict inequality between each pair of endpoints, $A > D$ holds and $B > D$ holds ($B \geq C > D$ gives $B > D$). Both follow. The trap is thinking that a $\geq$ link breaks the strict conclusion - it does not, as long as one strict $>$ appears in the chain.

5.14 Non-Verbal Reasoning

5.14.1 Mirror and water images

Mirror image - LEFT and RIGHT are interchanged (vertical flip axis)

Letters UNCHANGED in a mirror (when the mirror is vertical, beside the text):

A H I M O T U V W X Y

These are symmetric about a VERTICAL axis.

Water image - TOP and BOTTOM are interchanged (horizontal flip axis)

Letters UNCHANGED in water:

B C D E H I K O X

These are symmetric about a HORIZONTAL axis.

Numbers unchanged in water: 0 1 3 8

Q. [PYQ-TYPE] Which letters remain unchanged in their mirror image?

- (a) B, C, D
- (b) A, H, I, M
- (c) E, F, G
- (d) P, Q, R

Why: Mirror images preserve letters that are symmetric about a **vertical** axis: A, H, I, M, O, T, U, V, W, X, Y. B, C, D and E are symmetric about a ***horizontal*** axis, so they survive a **water** image instead - a classic swap in the options.

5.14.2 Paper folding and cutting

Method Work **backwards**. Start from the final cut shape and unfold one step at a time, mirroring the holes across each fold line. Each unfolding **doubles** the number of holes.

Key fact A paper folded n times and cut once produces 2^n holes when opened, provided the cut does not lie on a fold line.

5.14.3 Figure series and analogy

What to check	Question to ask
Rotation	Is the figure turning? By how much - 45, 90, 180 degrees? Clockwise or anticlockwise?
Reflection	Is it flipping across an axis?
Addition or removal of elements	Are lines, dots or shapes being added each step?
Movement	Is an element moving position by a fixed rule (one corner clockwise each step)?
Shading	Is the shading shifting or inverting?
Counting	Is the number of sides, lines or dots following a series?

The order to check figure series

Check in this sequence: **rotation, then movement of a marker, then addition/subtraction of elements, then shading**. Roughly 70% of figure series in these papers are simple rotation or a marker moving one position per step.

5.15 Miscellaneous Types

5.15.1 Dice and cubes

Standard dice rule On a standard die, **opposite faces sum to 7**: 1-6, 2-5, 3-4.

Two views of a die If two faces are **common** to both views, the remaining two faces are opposite each other.

Painted cube A cube painted on all sides and cut into n^3 smaller cubes gives: **8** cubes with 3 painted faces (the corners), **$12(n-2)$** with 2 faces (the edges), **$6(n-2)^2$** with 1 face (the centres), and **$(n-2)^3$** with **0** painted faces (the interior).

Q. [EXPECTED] [NUMERICAL] A cube painted on all faces is cut into 27 equal smaller cubes. How many have no face painted?

- (a) 8
- (b) 4
- (c) **1**
- (d) 6

Why: Here $n = 3$, so cubes with no painted face = $(n-2)^3 = 1^3 = 1$ - only the very centre cube. Check the rest: corners 8, edges $12(1) = 12$, faces $6(1) = 6$, interior 1. Total $8+12+6+1 = 27$. Correct.

5.15.2 Logical order of words

Arrange in a natural sequence - by size, by time, by process order, or by hierarchy.

Q. [PYQ-TYPE] Arrange in a logical order: stomach, mouth, food pipe, large intestine, small intestine

- (a) **mouth, food pipe, stomach, small intestine, large intestine**
- (b) mouth, stomach, food pipe, small intestine, large intestine
- (c) food pipe, mouth, stomach, large intestine, small intestine
- (d) mouth, food pipe, stomach, large intestine, small intestine

Why: Follow the actual path of food through the digestive system: **mouth, food pipe (oesophagus), stomach, small intestine, large intestine**. Note the last two - the **small** intestine comes before the large one, which is the point of the question.

Q. [EXPECTED] Arrange in logical order: seed, plant, flower, fruit, tree

- (a) **seed, plant, tree, flower, fruit**
- (b) seed, plant, flower, fruit, tree
- (c) plant, seed, tree, flower, fruit
- (d) seed, tree, plant, flower, fruit

Why: The developmental sequence is seed, then a young plant, which grows into a tree, which bears flowers, which become fruit: **seed, plant, tree, flower, fruit**.

5.15.3 Statement and conclusion / assumption

Statement-Conclusion Accept the statement as true. A conclusion follows only if it is **necessarily** implied - not merely plausible or likely.

Statement-Assumption An assumption is something **taken for granted** for the statement to make sense. Test it by negation: if denying the assumption destroys the statement, it is a valid assumption.

Common trap Anything requiring **outside knowledge** or an extra leap is **not** a valid conclusion, however true it may be in reality.

Q. [EXPECTED] [TRAP] Statement: "Use our shampoo for shining, healthy hair." Which assumption is implicit?

- (a) All shampoos are harmful
- (b) **People want shining, healthy hair**
- (c) No other shampoo works
- (d) Hair problems are common

Why: The advertisement only makes sense if people **desire** shining, healthy hair - deny that and the whole message collapses, so it is the implicit assumption. The other options are far stronger claims the statement does not need.

The correct assumption is always the minimum needed, never the boldest.

5.16 Reasoning Revision Sheet

5.16.1 Every formula in one place

[FORMULA] The complete reasoning formula sheet

ALPHABET

EJOTY anchors: E=5, J=10, O=15, T=20, Y=25

Reverse position of a letter = $27 - (\text{its position})$

RANKING

Total = (rank from top) + (rank from bottom) - 1

Rank from bottom = Total - (rank from top) + 1

Persons between = (difference of positions from same end) - 1

DIRECTIONS

Shortest distance across a right angle = $\sqrt{a^2 + b^2}$

Triples: 3-4-5, 5-12-13, 8-15-17, 7-24-25, 9-40-41

Facing SOUTH reverses left and right

CLOCK

Angle = $|30H - 5.5M|$ (subtract from 360 if over 180)

Minute hand 6 deg/min, hour hand 0.5 deg/min

Hands coincide 22 times a day, opposite 22 times, right angle 44 times

CALENDAR

Ordinary year = 1 odd day; Leap year = 2 odd days

Leap: divisible by 4, but century must be divisible by 400

100 yr = 5 odd days, 200 = 3, 300 = 1, 400 = 0

0=Sunday, 1=Monday, ... 6=Saturday

DICE AND CUBES

Opposite faces of a standard die sum to 7

Painted cube cut into n^3 :

3 faces painted = 8

2 faces painted = $12(n-2)$

1 face painted = $6(n-2)^2$

0 faces painted = $(n-2)^3$

MIRROR AND WATER

Mirror-safe letters: A H I M O T U V W X Y

Water-safe letters : B C D E H I K O X

Water-safe digits : 0 1 3 8

SYLLOGISM

All + All -> All

Some + All -> Some

All + Some -> nothing

"Some" reverses; "All" does NOT reverse

5.16.2 The five habits that produce 19-20 out of 20

- 1. Convert letters to numbers immediately.** Every letter-series and coding question becomes arithmetic the moment you apply EJOTY.
- 2. Draw, do not imagine.** Blood relations, directions, seating and syllogisms are all solved on paper. Candidates who attempt them mentally lose 4-5 marks per paper.
- 3. Start puzzles from the most definite statement,** never from the first one.
- 4. Verify at the end.** Once you have an arrangement, re-read every statement to confirm it fits. This catches almost every error before it costs you.
- 5. Never exceed 2 minutes on one question.** Flag and return. A puzzle you cannot crack in 2 minutes is worth the same 1 mark as a synonym you can answer in 8 seconds.

PART 6

TEST OF ENGLISH LANGUAGE AND COMPREHENSION

20 marks. Grammar and error-spotting are rule-based and fully learnable. Vocabulary is the only open-ended part, and this Part gives you the highest-yield lists. Target 17-19.

6.1 What This Section Actually Contains

6.1.1 The breakdown

Topic	Typical questions	Nature
Spotting errors / sentence correction	3-4	Pure rules - fully learnable
Fill in the blanks (grammar or vocabulary)	2-3	Rules plus word sense
Synonyms	2-3	Vocabulary
Antonyms	2-3	Vocabulary
Idioms and phrases	2-3	Memorisation
One-word substitution	1-2	Memorisation
Spelling correction	1-2	Memorisation
Active and passive voice	1-2	Pure rules
Direct and indirect speech	1	Pure rules
Sentence rearrangement	1-2	Technique
Reading comprehension	4-5	Technique

Roughly 11-13 of the 20 marks are RULE-BASED - grammar, voice, narration, comprehension technique. Those are the marks to secure first. Vocabulary is the remaining 7-9, and it rewards steady daily accumulation, not last-minute cramming.

6.2 Grammar Rules That Are Actually Tested

6.2.1 Subject-verb agreement - the highest-yield rule set

Rule	Correct usage
Singular subject takes a singular verb	The boy plays .
Two subjects joined by 'and' take a plural verb	Ram and Shyam are here.
But if 'and' joins one idea, use singular	Bread and butter is my breakfast.
Subjects joined by or / nor / either-or / neither-nor: verb agrees with the NEAREST subject	Neither the students nor the teacher was present.
'Each', 'every', 'either', 'neither', 'everyone', 'nobody', 'anyone' are always SINGULAR	Each of the boys is present.
'One of' + plural noun + singular verb	One of my friends is a doctor.
Collective nouns take singular when acting as one unit	The team is playing well.
Uncountable nouns are singular	The information is useful.
'The number of' takes singular; 'a number of' takes plural	The number of students is rising. A number of students are absent.
Distance, time, money, weight as one quantity take singular	Ten kilometres is a long walk. Fifty rupees is not enough.
'Police', 'people', 'cattle', 'gentry' are always PLURAL	The police have arrived.
'News', 'mathematics', 'physics', 'economics', 'politics' end in -s but are SINGULAR	The news is good. Mathematics is difficult.
'A pair of' takes singular	A pair of shoes is missing.

Q. [PYQ-TYPE] Spot the error: "Each of the students / have submitted / their assignment / on time."

- (a) Each of the students
- (b) **have submitted**
- (c) their assignment
- (d) on time

Why: "Each" is always **singular**, so it needs "**has** submitted", not "have". This is the single most frequently tested rule in the section. Related singular words: each, every, either, neither, everyone, everybody, anyone, nobody, someone, none.

Q. [PYQ-TYPE] Choose the correct sentence.

- (a) Neither of the boys were present.
- (b) **Neither of the boys was present.**
- (c) Neither of the boys are present.
- (d) Neither of the boys have present.

Why: "Neither" is singular, so it takes "**was**". The plural "boys" immediately before the verb is the deliberate distraction - the verb agrees with **neither**, not with boys.

Q. [EXPECTED] [TRAP] Choose the correct sentence.

- (a) The number of accidents are increasing.
- (b) **The number of accidents is increasing.**
- (c) A number of accidents is increasing.
- (d) Numbers of accidents is increasing.

Why: "**The number of**" is a single quantity, hence **singular** ("is"). But "**a number of**" means "several" and is **plural** ("A number of accidents **are** reported daily"). Learn both together - examiners test the pair.

6.2.2 Tenses

Rule	Example
Universal truths and habitual actions take simple present	The sun rises in the east.
'Since' and 'for' with a period take present perfect continuous	He has been living here since 2010.
After 'if' in a conditional, do not use 'will'	If it rains , we will stay home. (NOT "if it will rain")
Two past actions: the earlier takes past perfect	The train had left before I reached.
'Yesterday', 'ago', 'last night' need simple past	He went yesterday. (NOT "has gone yesterday")
After 'wish' and 'if only', use past subjunctive	I wish I were rich. (NOT "was")
'It is time' takes past tense	It is time we left .
After 'as if' / 'as though', use past	He behaves as if he were the boss.

Q. [PYQ-TYPE] Fill in the blank: "He ___ in this city since 2015."

- (a) lives
- (b) **has been living**
- (c) is living
- (d) lived

Why: "Since" with a starting point requires the **present perfect continuous** for an action still going on. Remember: **since** takes a point in time (since Monday), **for** takes a duration (for three years).

Q. [EXPECTED] [TRAP] Choose the correct sentence.

- (a) If he will come, I will meet him.
- (b) **If he comes, I will meet him.**
- (c) If he came, I will meet him.
- (d) If he come, I will meet him.

Why: In a first-conditional sentence, the **if-clause takes the simple present** and the main clause takes "will". Putting "will" in the if-clause is one of the most common errors tested.

6.2.3 Articles

Rule	Example
'A' before a consonant SOUND	a book, a university (yoo sound), a one-rupee note (wun sound)
'An' before a vowel SOUND	an apple, an hour (silent h), an MP (em sound), an honest man
'The' with unique things	the sun, the moon, the Earth
'The' with rivers, seas, mountain ranges, groups of islands	the Ganga, the Himalayas
NO 'the' with single mountains, lakes, continents, countries (usually)	Mount Everest, Lake Chilka, Asia, India
'The' with superlatives and ordinals	the best, the first
'The' before an adjective used as a noun for a class	the poor, the rich
No article before languages, meals, games	He speaks English. We had lunch. He plays cricket.
'The' with the names of newspapers, ships, holy books	The Hindu, The Ramayana

Q. [PYQ-TYPE] [TRAP] Fill in the blank: "He is ___ honest man."

- (a) a
- (b) **an**
- (c) the
- (d) no article

Why: "Honest" begins with a **silent h**, so the sound is a vowel - hence "**an**". The rule depends on **sound, not spelling**. Compare "a hotel" (h pronounced) and "an hour" (h silent); also "a university" (yoo) but "**an** umbrella" (uh).

Q. [EXPECTED] Fill in the blank: "___ Ganga is a sacred river."

- (a) A
- (b) An
- (c) **The**
- (d) No article

Why: Rivers always take "the". So do seas, oceans, mountain ranges, deserts and groups of islands. Single mountains do not - "Mount Everest", never "the Mount Everest".

6.2.4 Prepositions

Depend	on	It depends on you.
Consist	of	The book consists of ten chapters.
Comprise	no preposition	The book comprises ten chapters. (NOT "comprises of")
Married	to	She is married to a doctor.
Congratulate	on	I congratulate you on your success.
Die	of (disease) / from (injury)	He died of cancer.
Superior / Inferior	to	He is superior to me. (NOT "than")
Prefer	to	I prefer tea to coffee. (NOT "than")
Different	from	This is different from that.
Suffer	from	She suffers from fever.
Insist	on	He insisted on going.
Deprive	of	He was deprived of his rights.
Accuse	of	He was accused of theft.
Charge	with	He was charged with murder.
Familiar	with (a thing) / to (a person)	I am familiar with the topic.
Look forward	to + verb-ing	I look forward to meeting you.

Q. [PYQ-TYPE] [TRAP] Fill in the blank: "He is superior ___ me in intelligence."

- (a) than
- (b) **to**
- (c) from
- (d) with

Why: Latin comparatives - superior, inferior, senior, junior, prior, posterior, prefer - all take "to", never "than". "He is senior to me", "I prefer tea to coffee".

Q. [EXPECTED] [TRAP] Choose the correct sentence.

- (a) The committee comprises of ten members.
- (b) **The committee comprises ten members.**
- (c) The committee is comprised with ten members.
- (d) The committee comprise of ten members.

Why: "Comprise" takes no preposition. Use "consists of" or "is composed of", but "comprises" alone. This is a favourite error to plant.

6.2.5 Active and passive voice

Voice conversion rules

ACTIVE : Subject + Verb + Object

PASSIVE: Object + form of BE + past participle + by + Subject

Active : Ram writes a letter.

Passive: A letter is written by Ram.

TENSE MAPPING

Simple present	: is / am / are + V3	-> is written
Present continuous	: is / am / are being + V3	-> is being written
Present perfect	: has / have been + V3	-> has been written
Simple past	: was / were + V3	-> was written
Past continuous	: was / were being + V3	-> was being written
Past perfect	: had been + V3	-> had been written
Simple future	: will be + V3	-> will be written
Future perfect	: will have been + V3	-> will have been written
Modals	: can/may/must be + V3	-> can be written

IMPERATIVE sentences become "Let ..."

Active : Open the door.

Passive: Let the door be opened.

Q. [PYQ-TYPE] Change to passive voice: "The teacher punished the student."

- (a) The student punished by the teacher.
- (b) The student was punished by the teacher.**
- (c) The student is punished by the teacher.
- (d) The student has punished by the teacher.

Why: Simple past active becomes **was/were + past participle**: "was punished". Keeping the tense correct is what the question tests.

Q. [EXPECTED] Change to passive: "Someone has stolen my bag."

- (a) My bag has stolen.
- (b) My bag has been stolen.**
- (c) My bag is stolen.
- (d) My bag was stolen.

Why: Present perfect active becomes **has/have been + V3**. Note that when the subject is vague ("someone"), the "by" phrase is normally dropped.

6.2.6 Direct and indirect speech

Narration rules

- REPORTING VERB in past -> shift the reported tense one step back:
 present -> past ; past -> past perfect ; will -> would
 can -> could ; may -> might ; must -> had to
 (No shift if the reporting verb is in the present, or if the statement is a universal truth.)
- PRONOUNS change according to the speaker and listener.
- TIME AND PLACE words shift:
 now -> then today -> that day
 tomorrow -> the next day yesterday -> the previous day
 here -> there this -> that ago -> before
- CONNECTOR by sentence type:
 Statement -> that
 Question (wh-) -> the same wh-word
 Question (yes/no) -> if / whether
 Command / request -> to / not to (verb: ordered, requested, advised)
 Exclamation -> that, with the reporting verb 'exclaimed'

Q. [PYQ-TYPE] Convert: He said, "I am busy."

- (a) He said that I am busy.
- (b) He said that he was busy.**
- (c) He says that he is busy.
- (d) He said that he is busy.

Why: Two changes: the pronoun "I" becomes **"he"** (referring to the speaker), and the present "am" shifts back to **"was"** because the reporting verb "said" is past.

Q. [EXPECTED] Convert: She said, "Where is my book?"

- (a) She asked where is my book.
- (b) **She asked where her book was.**
- (c) She said where her book was.
- (d) She asked where was her book.

Why: For a **wh-question**, keep the wh-word as the connector, change the reporting verb to "**asked**", shift the pronoun ("my" to "her"), and convert to **statement word order** ("her book was", not "was her book").

6.3 Vocabulary

6.3.1 High-frequency synonyms

Word	Meaning	Synonyms
Abundant	Plentiful	Ample, copious, profuse
Adamant	Unyielding	Inflexible, obstinate, resolute
Alleviate	To relieve	Ease, mitigate, lessen
Ambiguous	Unclear, open to two readings	Vague, equivocal
Benevolent	Kind, generous	Benign, charitable, humane
Candid	Frank	Honest, straightforward, forthright
Complacent	Self-satisfied	Smug, unconcerned
Conceal	To hide	Cover, obscure, mask
Diligent	Hard-working	Industrious, assiduous
Eloquent	Fluent in speech	Articulate, expressive
Frugal	Careful with money	Thrifty, economical
Gregarious	Sociable	Outgoing, convivial
Hostile	Unfriendly	Antagonistic, belligerent
Impede	To obstruct	Hinder, hamper, obstruct
Lucid	Clear	Intelligible, transparent
Meticulous	Very careful about detail	Scrupulous, fastidious
Obsolete	Out of date	Outdated, archaic
Perpetual	Never-ending	Eternal, incessant
Prudent	Wise, cautious	Judicious, circumspect
Reluctant	Unwilling	Hesitant, averse, disinclined
Sluggish	Slow-moving	Lethargic, torpid, inactive
Tenacious	Persistent	Determined, dogged
Vindicate	To clear of blame	Exonerate, absolve, justify
Zealous	Very enthusiastic	Ardent, fervent, passionate

6.3.2 High-frequency antonyms

Word	Antonym
Abundant	Scarce
Amiable	Hostile
Barren	Fertile
Benevolent	Malevolent
Candid	Evasive
Concise	Verbose
Docile	Stubborn
Extravagant	Frugal
Frail	Robust

Word	Antonym
Genuine	Counterfeit
Humble	Arrogant
Innocent	Guilty
Lethargic	Energetic
Meagre	Plentiful
Novice	Expert
Optimist	Pessimist
Praise	Censure
Rigid	Flexible
Scarcity	Abundance
Transparent	Opaque
Unite	Divide
Vague	Precise
Wither	Flourish

6.3.3 Idioms and phrases

Idiom	Meaning
A blessing in disguise	Something that seems bad but turns out good
At a stone's throw	Very near
Beat about the bush	To avoid the main point
Bolt from the blue	A complete surprise, usually unpleasant
Break the ice	To start a conversation
Bury the hatchet	To make peace, end a quarrel
By leaps and bounds	Very rapidly
Call a spade a spade	To speak plainly
Cry over spilt milk	To regret what cannot be undone
Every inch	Completely, entirely
Get cold feet	To lose courage
Hit the nail on the head	To say exactly the right thing
In cold blood	Deliberately and without emotion
In the nick of time	Just in time
Let the cat out of the bag	To reveal a secret
Make both ends meet	To manage within one's income
Once in a blue moon	Very rarely
Play second fiddle	To take a subordinate role
Pull one's socks up	To make an extra effort
Rain cats and dogs	To rain heavily
Show the white feather	To act like a coward
Smell a rat	To suspect something wrong
Spill the beans	To disclose a secret

Idiom	Meaning
Take to task	To reprimand
Turn a deaf ear	To refuse to listen
Turn over a new leaf	To start afresh, reform
Under the weather	Feeling unwell
Win with flying colours	To succeed brilliantly

Q. [PYQ-TYPE] What does 'to beat about the bush' mean?

- (a) To work hard
- (b) **To avoid the main topic**
- (c) To search everywhere
- (d) To make noise

Why: The idiom means talking around a subject without addressing it directly. Note the trap "to search everywhere" - a literal reading of the words, which is exactly how idiom questions are built.

Q. [EXPECTED] 'A bolt from the blue' means:

- (a) A loud noise
- (b) A useful tool
- (c) **A sudden and unexpected event**
- (d) A bright light

Why: It refers to lightning from a clear sky - hence a complete and usually unwelcome surprise.

6.3.4 One-word substitution

Definition	Word
One who loves books	Bibliophile
One who studies birds	Ornithologist
One who collects coins	Numismatist
One who collects stamps	Philatelist
One who eats everything	Omnivorous
One who eats only vegetables	Vegetarian
One who cannot be corrected	Incorrigible
One who cannot read or write	Illiterate
One who talks too much	Loquacious / Garrulous
One who hates women	Misogynist
One who hates mankind	Misanthrope
One who lives alone	Recluse / Hermit
One who walks in sleep	Somnambulist
One who does not believe in God	Atheist
One who believes in God	Theist
Government by the people	Democracy
Government by one person	Autocracy / Monarchy
Government by officials	Bureaucracy
Government by the rich	Plutocracy
A speech made without preparation	Extempore
Words written on a tomb	Epitaph
A life history written by oneself	Autobiography

Definition	Word
A life history written by another	Biography
Medicine that counteracts poison	Antidote
A place where money is coined	Mint
A place where birds are kept	Aviary
A place where bees are kept	Apiary
A place for the preservation of animals	Zoo / Menagerie
Fear of water	Hydrophobia
Fear of foreigners	Xenophobia

6.3.5 Commonly confused pairs

Pair	Distinction
Accept / Except	Accept = to receive; Except = excluding
Affect / Effect	Affect = to influence (verb); Effect = the result (noun)
Advice / Advise	Advice = noun; Advise = verb
Principal / Principle	Principal = head of a school, or chief; Principle = a rule
Stationary / Stationery	Stationary = not moving; Stationery = writing materials
Complement / Compliment	Complement = to complete; Compliment = praise
Loose / Lose	Loose = not tight; Lose = to misplace
Its / It's	Its = possessive; It's = it is
Their / There / They're	Possessive / place / they are
Weather / Whether	Weather = climate; Whether = if
Cite / Site / Sight	Quote / place / vision
Council / Counsel	Council = an assembly; Counsel = advice or a lawyer
Beside / Besides	Beside = next to; Besides = in addition to
Farther / Further	Farther = physical distance; Further = additional extent

6.4 Sentence Rearrangement and Comprehension

6.4.1 Rearranging sentences

THE METHOD

- Step 1: Find the OPENING sentence. It introduces the subject with a full noun, not a pronoun, and depends on nothing before it.
- Step 2: Find the CLOSING sentence - a conclusion, result or summary.
- Step 3: Use LINK WORDS to chain the middle:
however, therefore, but, moreover, thus, consequently,
on the other hand, finally
- Step 4: Track PRONOUNS. A sentence beginning "He/She/It/They/This" must follow the sentence naming that thing.
- Step 5: Follow CHRONOLOGY or logical cause-and-effect.
- Step 6: Read your final order once, straight through, to check it flows.

The fastest way into a rearrangement question

Do not try to order all four or five parts at once. Instead, find the **one sentence that cannot possibly be first** (because it starts with a pronoun or a connector) and the **one that must be first**. That usually eliminates two or three of the four options immediately.

6.4.2 Reading comprehension

THE METHOD - questions before passage

- Step 1: Read the QUESTIONS first, quickly. Now you know what to look for.
- Step 2: Skim the passage once for its overall subject and tone.
- Step 3: Answer the FACTUAL questions first by locating the exact line.
- Step 4: Answer inference and tone questions last.
- Step 5: For vocabulary-in-context questions, substitute each option back into the original sentence and see which preserves the meaning.

GOLDEN RULE: the answer must come from the PASSAGE, never from your own knowledge of the world. If an option is true in real life but is not stated or implied in the passage, it is **WRONG**.

Question type	How to handle it
Direct / factual	Locate the line. The answer is stated
Vocabulary in context	Substitute each option into the sentence
Inference	Must follow necessarily from the text, not merely be plausible
Main idea / title	Should cover the whole passage, not one paragraph
Tone / attitude	Look at adjectives and adverbs the writer chooses
"The author would agree with"	Find explicit support in the text

The two traps in every comprehension set

Trap 1: the too-broad or too-narrow title. For "best title" questions, reject options that describe only one paragraph, and reject options broader than the passage.

Trap 2: the true-but-irrelevant option. Examiners include statements that are factually correct in the real world but absent from the passage. Your only authority is the passage.

6.5 English Revision Sheet

6.5.1 The twenty rules that carry the marks

The rules worth memorising verbatim

1. Each / every / either / neither / everyone / nobody = SINGULAR.
2. 'One of' + plural noun + SINGULAR verb.
3. Or / nor / either-or / neither-nor: verb agrees with the NEAREST subject.
4. 'The number of' = singular; 'a number of' = plural.
5. News, mathematics, physics, economics, politics = SINGULAR.
6. Police, people, cattle, gentry = PLURAL.
7. Superior, inferior, senior, junior, prior, prefer + TO (never 'than').
8. 'Comprise' takes NO preposition.
9. Article choice depends on SOUND, not spelling (an hour, a university).
10. Rivers, seas, mountain ranges, holy books, newspapers take THE.
11. Single mountains, lakes, continents take NO article.
12. In an if-clause use the simple present, never 'will'.
13. 'Since' + point in time; 'for' + duration.
14. Two past actions: the earlier one takes past perfect (had + V3).
15. After 'wish', 'as if', 'it is time': use the PAST form (I wish I were).
16. Passive = form of BE + past participle (V3) + by.
17. Imperative in passive becomes 'Let ... be ...'.
18. Reporting verb in past shifts the reported tense one step back.
19. Yes/no question in indirect speech takes 'if' or 'whether'.
20. In comprehension, the passage is the ONLY authority.

6.5.2 Habits that produce 17-19 out of 20

1. **Learn ten new words a day, in sentences.** Isolated word lists decay; words in context stick. Six weeks of this covers most of what these papers ask.
2. **Do error-spotting daily.** It is the most rule-dense question type, so it improves fastest with practice.
3. **When spotting errors, check in a fixed order:** subject-verb agreement, then tense, then preposition, then article, then pronoun. A systematic scan catches errors that reading "by feel" misses.
4. **Read the comprehension questions before the passage.** It halves the time you spend re-reading.
5. **Never answer comprehension from general knowledge.** If it is not in the passage, it is not the answer - regardless of whether it is true.

PART 7

TEST OF HINDI LANGUAGE AND COMPREHENSION

20 marks. Vyakaran is rule-based and highly scoring. Read Chapter 7.1 first - it explains an important technical limitation of this PDF and where to get the Devanagari version.

7.1 Important - Read Before You Study This Part

7.1.1 A technical limitation, stated plainly

This PDF was generated by a program I wrote from scratch, because no PDF library could be installed in the environment where the book was built. That generator uses the standard PDF fonts, which contain **Latin characters only**.

It therefore cannot print Devanagari. Not badly - at all.

I also could not solve this by embedding a Devanagari font, for two reasons: no Devanagari font file existed on the machine, and there was no network access to obtain one. Even with a font, correct Devanagari needs complex-script shaping - reordering of the **i** matra, formation of conjuncts like **ksha** and **tra** - which is a substantial piece of typographic engineering that I could not implement reliably enough to stake your preparation on.

So here is what I did instead - and where to get proper Hindi

In this PDF, Part 7 teaches Hindi grammar using **Roman transliteration plus English explanation**. This works well for *learning the rules* - the rule that a **visarga sandhi** joins a word ending in **ah** with a following consonant is fully learnable in transliteration.

For the Devanagari original, the same content is provided as a separate file alongside this book:

``hindi_section_devanagari.html`` - open it in any web browser. Devanagari renders perfectly, because the browser does the shaping. Then press **Ctrl+P** and choose **"Save as PDF"** to get a proper Hindi PDF with correct script.

``hindi_section_devanagari.md`` - the same content in Markdown, which renders Devanagari directly on GitHub if you are reading it there.

I would rather tell you this clearly than silently print boxes or question marks where the Hindi should be. **Use the HTML file for your actual Hindi preparation**, and use this Part to understand the rules in English first - which most candidates find genuinely easier as a starting point.

7.1.2 What this section contains

Topic (Hindi term)	English	Typical questions
Sandhi	Joining of sounds	1-2
Samas	Compound words	1-2
Upsarg and Pratyay	Prefixes and suffixes	1-2
Paryayvachi	Synonyms	2-3
Vilom	Antonyms	2-3
Anekarthi shabd	Words with multiple meanings	0-1
Muhavare and Lokoktiyan	Idioms and proverbs	2-3
Ashuddhi shodhan	Error correction	1-2
Vakya ke bhed	Types of sentences	1
Ling, Vachan, Karak	Gender, number, case	1-2
Tatsam-Tadbhav	Sanskrit-derived and colloquial forms	0-1
Ek shabd	One-word substitution	1
Apathit gadyansh	Unseen passage comprehension	4-5

As in English, roughly half of these marks are RULE-BASED - sandhi, samas, karak, error correction, sentence types. Those are the marks to secure first.

7.2 Sandhi - Joining of Sounds

7.2.1 The concept

Sandhi means the merging of the last sound of one word with the first sound of the next, producing a changed form. There are three main types.

Type	Rule	Example (transliterated)
Swar Sandhi (vowel)	Vowel meets vowel	vidya + arthi = vidyarthi
Vyanjan Sandhi (consonant)	Consonant meets vowel or consonant	sat + jan = sajjan
Visarg Sandhi	A word ending in the visarg (ah) meets another sound	manah + yog = manoyog

7.2.2 Swar Sandhi - the five sub-types

Sub-type	Rule	Example
Dirgh	Same vowels combine into their long form (a + a = aa, i + i = ee, u + u = oo)	dharm + arth = dharmarth ; kavi + indra = kavindra ; bhanu + uday = bhanuday
Guna	a + i = e ; a + u = o ; a + ri = ar	nar + indra = narendra ; sur + ish = suresh ; maha + rishi = maharshi
Vriddhi	a + e = ai ; a + o = au	ek + ek = aikya ; jal + oormi = jalaurmi
Yan	i + dissimilar vowel becomes y ; u becomes v	ati + adhik = atyadhik ; su + agat = swagat
Ayadi	e + a = ay ; ai + a = aay ; o + a = av ; au + a = aav	ne + an = nayan ; pau + an = pavan

Q. [PYQ-TYPE] "Vidyarthi" is formed by which sandhi?

- (a) Guna
- (b) Vriddhi
- (c) **Dirgh**
- (d) Yan

Why: Break it: **vidya + arthi**. Here **aa** meets **a**, two similar vowels merging into a long **aa** - which is **Dirgh sandhi**. Rule to remember: when the two vowels are of the **same family**, it is always Dirgh.

Q. [PYQ-TYPE] "Narendra" is an example of:

- (a) Dirgh
- (b) **Guna**
- (c) Vriddhi
- (d) Ayadi

Why: **nar + indra**. Here **a + i = e**, which is the defining pattern of **Guna sandhi**. Learn the three Guna combinations as a set: **a+i=e, a+u=o, a+ri=ar**.

Q. [EXPECTED] "Sajjan" is formed by which type of sandhi?

- (a) Swar sandhi
- (b) **Vyanjan sandhi**
- (c) Visarg sandhi
- (d) None

Why: **sat + jan = sajjan**. A **consonant** (t) meets a consonant (j) and changes, so this is **Vyanjan sandhi**. If either side of the join involves a consonant transformation, it is Vyanjan, not Swar.

Q. [EXPECTED] "Manoyog" involves which sandhi?

- (a) Dirgh
- (b) Guna
- (c) **Visarg**
- (d) Yan

Why: **manah + yog**. The first word ends in the **visarg (ah)**, which is the signature of **Visarg sandhi**. Whenever you see **ah / uh** at the join, check Visarg first.

7.3 Samas - Compound Words

7.3.1 The six types

Samas is the combining of two or more words into one, dropping the connecting words. The type is decided by **which part carries the main meaning**.

Type	Rule - which part dominates	Example (transliterated)
Avyayibhava	The first word dominates; the compound acts as an adverb. Often begins with prati, anu, bhar, yatha, aa	pratidin (every day), yathashakti (according to ability), bharpet (a full stomach)
Tatpurush	The second word dominates; a case-ending is dropped	rajkumar (raja ka kumar), dharmagranth (dharma ka granth), gyaneshwar
Karmdharaya	One word is an adjective describing the other; both refer to the same thing	neelkamal (blue lotus), mahadev (great god), peetambar
Dvigu	The first word is a number ; denotes a collection	navratna (nine gems), chaturbhuj (four-armed), trilok (three worlds)
Dvandva	Both words are equally important; an 'and' is implied	mata-pita (mother and father), raat-din (night and day), dukh-sukh
Bahuvrihi	Neither word dominates; the compound denotes a third thing altogether	dashanan (one with ten faces, i.e. Ravana), lambodar (one with a large belly, i.e. Ganesha), chakradhar

How to identify the samas type in ten seconds

Ask the questions in this order: 1. Does it start with **prati / anu / yatha / bhar / aa** and act as an adverb? Then **Avyayibhav**. 2. Does the first word denote a **number**? Then **Dvigu**. 3. Do **both** words matter equally, with an implied "and"? Then **Dvandva**. 4. Does the compound refer to a **third person or thing** not named in either word? Then **Bahuvrihi**. 5. Is one word an **adjective** of the other, both pointing to the same thing? Then **Karmdharaya**. 6. Otherwise, if the **second** word dominates, it is **Tatpurush**.

Q. [PYQ-TYPE] "Dashanan" is an example of which samas?

- (a) Dvigu
- (b) Karmdharaya
- (c) Tatpurush
- (d) **Bahuvrihi**

Why: "Dashanan" literally means "ten faces", but it actually denotes **Ravana** - a **third entity** named by neither component. That is the definition of **Bahuvrihi**. The trap is choosing Dvigu because "dash" is a number - but in Dvigu the compound denotes the *collection itself* (as in navratna, the nine gems), not a third thing.

Q. [PYQ-TYPE] "Mata-pita" belongs to which samas?

- (a) Tatpurush
- (b) **Dvandva**
- (c) Dvigu
- (d) Bahuvrihi

Why: Both words are equally important and an **"and"** is implied between them, which is **Dvandva**. Such compounds are usually written with a hyphen - a useful visual clue.

Q. [EXPECTED] "Pratidin" is an example of:

- (a) Tatpurush
- (b) Dvigu
- (c) **Avyayibhav**
- (d) Karmdharaya

Why: It begins with the prefix **prati** and functions as an **adverb** ("every day"), which marks **Avyayibhav samas**.

Q. [EXPECTED] "Neelkamal" is which samas?

- (a) Dvandva
- (b) **Bahuvrihi**

(c) **Karmdharaya**

(d) Dvigu

Why: "Neel" (blue) is an **adjective** describing "kamal" (lotus), and both words refer to the **same object**. That is **Karmdharaya**.

7.4 Upsarg and Pratyay - Prefixes and Suffixes

Upsarg (prefix)	Meaning	Example
pra	forward, excess	prabal, pragati
apa	away, bad	apman, apvyay
sam	together	samyog, sammelan
ni	down, without	nivas, nishedh
nir / nis	without	nirdhan, nishchay
dur / dus	bad, difficult	durgun, dushkarm
vi	special, apart	vishesh, viyog
adhi	above, extra	adhikar, adhyaksh
anu	after, following	anudar, anuvad
ati	excess	atyant, atirek
a / an	not	abhav, anpadh (negation)
su	good	sugam, sushil
prati	towards, opposite	pratidin, prativad
upa	near, subordinate	upvan, upadhyaksh

Pratyay (suffix)	Function	Example
-ta	forms an abstract noun	manavta, sundarta
-pan	forms an abstract noun	bachpan, ladakpan
-vala	denotes a possessor or doer	gaadivala, dhanvala
-kar	denotes a doer	kalakar, chitrakar
-ik	forms an adjective	samajik, dharmik
-hin	denotes lacking	balhin, dhanhin
-mat / -van	denotes possessing	buddhimat, dhanvan
-i / -in	denotes belonging	desi, gyani

Krit pratyay A suffix added to a **verb root** - e.g. likh + ak = lekhak (writer).

Tadhit pratyay A suffix added to a **noun, pronoun or adjective** - e.g. manav + ta = manavta.

Q. [EXPECTED] In the word "nirdhan", the prefix is:

- (a) nird
- (b) **nir**
- (c) dhan
- (d) an

Why: **nir** is the prefix (meaning "without") and **dhan** (wealth) is the root, so nirdhan means "without wealth", i.e. poor. Related prefixes carrying negation: **nir, nis, a, an, dur, apa**.

Q. [EXPECTED] "Manavta" is formed by adding which suffix?

- (a) -van
- (b) -ik
- (c) **-ta**
- (d) -pan

Why: **manav + ta**. The suffix **-ta** converts a noun into an **abstract noun** (humanity). It is a **tadhit pratyay** because it attaches to a noun rather than a verb root.

7.5 Karak, Ling and Vachan

7.5.1 Karak - the eight cases

Karak	English	Marker (vibhakti)
Karta	Subject / doer	ne
Karm	Object	ko
Karan	Instrument ("by means of")	se, ke dwara
Sampradan	Recipient ("for")	ke liye, ko
Apadan	Separation ("from")	se (in the sense of separation)
Sambandh	Possession	ka, ke, ki
Adhikaran	Location	mein, par
Sambodhan	Address / calling out	he, are, ho

The karak distinction that is always tested

Karan and **Apadan** both use the marker **se**, and telling them apart is the standard question.

Karan means "by means of" - an instrument. *Kalam se likha* (wrote **with** a pen). **Apadan** means "from" - separation or origin. *Ped se patta gira* (the leaf fell **from** the tree).

Ask: is the **se** doing the work (Karan), or marking a source you are moving away from (Apadan)?

7.5.2 Ling and Vachan

Ling (gender) Hindi has two: **Pulling** (masculine) and **Stri-ling** (feminine). Unlike English, inanimate objects also carry gender - *kitab* is feminine, *makan* is masculine.

Vachan (number) Two: **Ekvachan** (singular) and **Bahuvachan** (plural). Common transformations: ladka to ladke, kitab to kitaben, nadi to nadiyan, chidiya to chidiyan.

7.6 Vakya - Sentences and Error Correction

7.6.1 Types of sentence by structure

Type	Description
Saral vakya	Simple - one subject, one predicate, one finite verb
Sanyukt vakya	Compound - two independent clauses joined by aur, parantu, kintu, isliye
Mishra vakya	Complex - one main clause plus one or more subordinate clauses, joined by ki, jo, jab, yadi, kyonki

7.6.2 Types of sentence by meaning

Vidhanvachak (assertive), Nishedhvachak (negative), Prashnavachak (interrogative), Aagyavachak (imperative), Vismayadibodhak (exclamatory), Sandehvachak (doubtful), Ichchhavachak (optative), Sanketvachak (conditional).

7.6.3 Common errors tested in ashuddhi shodhan

Error type	Wrong to Right (transliterated)
Ling agreement	"Meri pustak achha hai" to "Meri pustak achhi hai" (pustak is feminine)
Vachan agreement	"Ve ladka hai" to "Ve ladke hain"
Wrong vibhakti	"Main school ko jata hun" to "Main school jata hun"
Redundancy (punarukti)	"Kripya aap baithiye" to "Kripya baithiye"
Wrong word order	Verb should come at the end in Hindi
Tatsam-tadbhav confusion	Using the wrong register of a word
Sandhi and spelling errors	"anusshasan" to "anushasan"

The two checks that catch most Hindi errors

Check 1 - gender agreement. Find the noun, decide its gender, then confirm every adjective and the verb agree with it. Most planted errors are gender mismatches, because English speakers do not naturally track the gender of inanimate objects.

Check 2 - the vibhakti. Verify that the case marker (ne, ko, se, mein, par) matches the karak the sentence actually needs. A wrong "ko" or "se" is the second most common planted error.

7.7 Vocabulary - Paryayvachi, Vilom, Muhavare

7.7.1 Paryayvachi (synonyms) - high-frequency sets

Word	Synonyms (transliterated)
Sooraj (sun)	Ravi, Bhaskar, Dinkar, Aditya, Surya, Prabhakar
Chandra (moon)	Chand, Shashi, Rajanish, Mayank, Nishakar, Sudhakar
Aag (fire)	Agni, Anal, Pavak, Jwala, Vahni
Paani (water)	Jal, Neer, Toy, Salil, Ambu, Vari
Aakash (sky)	Gagan, Nabh, Ambar, Vyom, Antariksh
Pruthvi (earth)	Dharti, Bhoomi, Dhara, Vasundhara, Achala
Hawa (air)	Vayu, Pavan, Samir, Anil, Marut
Phool (flower)	Pushp, Kusum, Sumang, Prasun
Nadi (river)	Sarita, Tatini, Nirjharini, Apaga
Samudra (sea)	Sagar, Ratnakar, Jaldhi, Ambudhi
Raja (king)	Nripati, Bhoopati, Narpati, Samrat
Ghoda (horse)	Ashwa, Turang, Haya, Saindhav
Kamal (lotus)	Pankaj, Jalaj, Neeraj, Sarasij, Aravind
Ped (tree)	Vruksh, Padap, Vitap, Rukh
Ma (mother)	Janani, Matr, Amba, Jananee

7.7.2 Vilom (antonyms) - high-frequency pairs

Word	Antonym
Aadi	Ant
Andhkar	Prakash
Aaman	Nirman (or Vinash for construction/destruction)
Ashtha	Anastha
Uday	Ast
Utkarsh	Apkarsh
Anurag	Virag
Aadar	Anadar
Gyan	Agyan
Jeevan	Mrityu
Nirdhan	Dhanvan
Sukh	Dukh
Swarg	Narak
Shubh	Ashubh
Saral	Kathin
Sajeev	Nirjeev
Ucch	Nimn
Yash	Apyash

Word	Antonym
Sadupyog	Durupyog
Krutagya	Krutghna

7.7.3 Muhavare (idioms) - high-frequency

Muhavara	Meaning
Aankh ka tara	Very dear one
Aankhon mein dhool jhonkna	To deceive
Naak katna	To be disgraced
Kaan bharna	To instigate against someone
Sir par chadhana	To pamper excessively
Haath malna	To regret
Peeth dikhana	To flee
Daant khatte karna	To defeat badly
Ghee ke diye jalana	To celebrate joyously
Chaar din ki chandni	Short-lived happiness
Loha lena	To confront strongly
Angoothha dikhana	To refuse at the last moment
Ghar ka bhedi	An insider who reveals secrets
Aag mein ghee daalna	To worsen a quarrel
Titri ka pet phatna	To be unable to keep a secret
Naukon chane chabana	To face great difficulty
Andhe ki laathi	The only support
Thali ka baingan	One with no fixed opinion

7.8 Apathit Gadyansh - Unseen Passage

7.8.1 The method

The comprehension technique is **identical to English** - only the language changes.

THE METHOD

Step 1: Read the QUESTIONS first, so you know what to look for.

Step 2: Read the passage once for the overall subject and tone.

Step 3: Answer factual questions by locating the exact line.

Step 4: Answer the "shirshak" (title) question LAST, after you have understood the whole passage.

Step 5: For a title, choose the option that covers the WHOLE passage - not one paragraph, and not something broader than the passage.

GOLDEN RULE: the answer must come from the PASSAGE. If an option is true in general knowledge but is not stated in the passage, it is wrong.

The title question in Hindi comprehension

"Upsyukt shirshak" (most appropriate title) questions almost always appear, and candidates lose them by choosing the most *interesting* option rather than the most *representative* one.

Test each option against two questions: does it cover the **entire** passage, and does it claim **more** than the passage says? The right answer covers everything and claims nothing extra.

7.9 Hindi Revision Sheet

7.9.1 The rules to memorise

Hindi rules worth memorising

SANDHI

Same vowel family joining -> DIRGH
 a + i = e ; a + u = o ; a + ri = ar -> GUNA
 a + e = ai ; a + o = au -> VRIDDHI
 i / u + dissimilar vowel becomes y / v -> YAN
 e, ai, o, au + vowel becomes ay, aay, av, aav -> AYADI
 Consonant transformation at the join -> VYANJAN
 First word ends in visarg (ah) -> VISARG

SAMAS

Adverb, starts with prati/anu/yatha/bhar -> AVYAYIBHAV
 First word is a NUMBER, denotes a collection -> DVIGU
 Both words equal, 'and' implied -> DVANDVA
 Denotes a THIRD thing not named -> BAHUVRIHI
 One word is an adjective of the other -> KARMDHARAYA
 Second word dominates -> TATPURUSH

KARAK MARKERS

ne (karta), ko (karm), se/ke द्वारा (karan),
 ke liye (sampradan), se-separation (apadan),
 ka/ke/ki (sambandh), mein/par (adikaran), he/are (sambodhan)

THE se DISTINCTION

"by means of" = KARAN ; "from / away from" = APADAN

ERROR CORRECTION - check in this order

1. Gender agreement (ling)
2. Number agreement (vachan)
3. Case marker (vibhakti)
4. Redundancy (punarukti dosh)
5. Word order - the verb goes last

7.9.2 Habits that produce 17-19 out of 20

1. **Use the Devanagari HTML file for actual practice.** This Part teaches the rules; the script must be practised in the script.
2. **Learn sandhi and samas by their identification rules,** not by memorising word lists. The rules generalise; lists do not.
3. **For error correction, always check gender first.** It is the most frequently planted error.
4. **Learn paryayvachi in family sets** - all the words for sun together, all for moon together. They are asked as sets.
5. **Do the Hindi section first in the exam.** For most candidates it is the fastest 20 marks in the paper, and banking it early settles the nerves.

PART 8

FINAL REVISION AND EXAM STRATEGY

Read only this Part in the last seven days. Everything here is either a formula you must recall instantly or a habit that converts knowledge into marks.

8.1 The Seven-Day Revision Plan

Day	Revise	Why
Day 7	Chapter 4.1 (all maths formulas) + Chapter 5.16 (all reasoning formulas)	These two chapters carry 40 of the 100 marks and are pure recall
Day 6	Chapter 6.5 (twenty English rules) + Chapter 7.9 (Hindi rules)	Another 40 marks, also rule-based
Day 5	Chapter 2.9 (forty static GK facts) + Chapter 2.8 (Delhi GK)	Delhi GK is your differentiator - almost nobody prepares it
Day 4	Your own current-affairs notebook, last three months	Only your own notes. Do not open a new source now
Day 3	One full-length timed mock in the exam's real order	Tests stamina and the section-order strategy, not knowledge
Day 2	Your leak list only - the mistakes you actually made	The highest-value document you own by this stage
Day 1	Chapter 8.2 and 8.3 of this Part. Then stop. Sleep	Recall collapses with fatigue, and this paper is a recall paper

What not to do in the last week

Do not start a new book, a new test series or a new current-affairs source. Do not attempt a topic you have never touched - a topic learned in the final week is not retained under exam pressure, and the hours are better spent making certain things certain.

Do not stay up late the night before. This exam is 200 questions in 120 minutes, decided almost entirely by **speed of recall**. Recall is the first cognitive function to degrade with sleep loss. A rested candidate who knows 80% beats a tired candidate who knows 90%.

8.2 The One-Page Formula Sheet

8.2.1 Mathematics

Everything you must recall instantly

PERCENTAGE

Percentage change = $(\text{change} / \text{ORIGINAL}) \times 100$
 Increase then decrease by $x\%$ $\rightarrow$ net DECREASE of $x^2/100\%$
 Successive changes $a\%$ and $b\%$ $\rightarrow a + b + ab/100$
 A is $x\%$ more than B $\rightarrow$ B is $100x/(100+x)\%$ less than A

PROFIT AND LOSS

Profit% on COST price ; Discount% on MARKED price
 Successive discounts $a\%$ and $b\%$ $\rightarrow a + b - ab/100$
 Same profit% and loss% at same SP $\rightarrow$ net LOSS of $x^2/100\%$
 Faulty weight gain% = $[\text{error} / (\text{true} - \text{error})] \times 100$

INTEREST

SI = $\text{PRT}/100$
 CI: $A = P(1 + R/100)^T$
 CI - SI for 2 years = $P(R/100)^2$
 Doubling time at SI = $100/R$ years

RATIO AND AVERAGE

Divide X in $a:b:c \rightarrow Xa/(a+b+c)$ etc.
 Mean proportional of a and b = $\sqrt{ab}$
 Average of first n naturals = $(n+1)/2$
 Sum of first n naturals = $n(n+1)/2$
 Replace one value: new avg = $\text{old} + (\text{new} - \text{old})/n$

TIME AND WORK

A and B together = $ab/(a+b)$ days
 Outlet pipe = NEGATIVE work
 $M_1 D_1 H_1 / W_1 = M_2 D_2 H_2 / W_2$

SPEED

km/hr to m/s: $\times 5/18$ m/s to km/hr: $\times 18/5$
 Equal DISTANCES at x and y $\rightarrow$ average = $2xy/(x+y)$
 Equal TIMES at x and y $\rightarrow$ average = $(x+y)/2$
 Train + platform = both lengths ; train + pole = train length only
 Same direction: subtract speeds ; opposite: add speeds
 Boat: speed = $(\text{down} + \text{up})/2$; stream = $(\text{down} - \text{up})/2$

MENSURATION

Square diagonal = $a\sqrt{2}$; Cube diagonal = $a\sqrt{3}$
 Circle: area πr^2 , circumference $2\pi r$
 Equilateral triangle area = $(\sqrt{3}/4) a^2$
 Cylinder volume = $\pi r^2 h$; Cone = $(1/3) \pi r^2 h$
 Sphere volume = $(4/3) \pi r^3$; surface = $4\pi r^2$
 Double the radius $\rightarrow$ area $\times 4$, volume $\times 8$

NUMBERS

HCF $\times$ LCM = product of the two numbers
 Same remainder r $\rightarrow$ answer = LCM + r
 Constant (divisor - remainder) = d $\rightarrow$ answer = LCM - d
 Divisibility by 11: (odd-position sum) - (even-position sum)

PERCENTAGE-FRACTION (learn cold)

$50\% = 1/2$ $33.33\% = 1/3$ $25\% = 1/4$ $20\% = 1/5$ $16.67\% = 1/6$
 $14.28\% = 1/7$ $12.5\% = 1/8$ $11.11\% = 1/9$ $10\% = 1/10$ $6.25\% = 1/16$

8.2.2 Reasoning

Everything you must recall instantly

ALPHABET	EJOTY: E=5 J=10 O=15 T=20 Y=25 Reverse position = 27 - position
RANKING	Total = top rank + bottom rank - 1 Between two = difference from same end - 1
DIRECTION	Shortest = $\sqrt{a^2 + b^2}$ Triples: 3-4-5, 5-12-13, 8-15-17, 7-24-25 Facing SOUTH reverses left and right
CLOCK	Angle = $ 30H - 5.5M $ (over 180? subtract from 360) Coincide 22 times a day ; right angles 44 times a day
CALENDAR	Ordinary year = 1 odd day ; leap = 2 odd days Leap: div by 4, century must be div by 400 0=Sunday ... 6=Saturday
DICE/CUBE	Opposite faces sum to 7 n^3 cut: 3 faces=8, 2 faces=12(n-2), 1 face=6(n-2)^2, 0 faces=(n-2)^3
MIRROR	Safe letters: A H I M O T U V W X Y
WATER	Safe letters: B C D E H I K O X ; digits 0 1 3 8
SYLLOGISM	All+All=All ; Some+All=Some ; All+Some=nothing "Some" reverses ; "All" does NOT reverse

8.2.3 English and Hindi

The rules that carry the language marks

ENGLISH

Each/every/either/neither/everyone/nobody = SINGULAR
 'One of' + plural noun + SINGULAR verb
 Or/nor: verb agrees with the NEAREST subject
 'The number of' singular ; 'a number of' plural
 News, mathematics, physics, economics = SINGULAR
 Police, people, cattle = PLURAL
 Superior/inferior/senior/junior/prefer + TO (never 'than')
 'Comprise' takes NO preposition
 Articles follow SOUND not spelling (an hour, a university)
 No 'will' in an if-clause
 'Since' + point in time ; 'for' + duration
 Earlier of two past actions -> past perfect
 After wish / as if / it is time -> use PAST (I wish I were)
 Passive = BE + past participle + by
 Reporting verb in past -> shift reported tense one step back
 Yes/no question in indirect speech -> if / whether
 Comprehension: the PASSAGE is the only authority

HINDI

Same vowel family -> DIRGH ; a+i=e -> GUNA ; visarg -> VISARG
 Number word + collection -> DVIGU
 Denotes a THIRD thing -> BAHUVRIHI
 Both equal, 'and' implied -> DVANDVA
 Adjective + noun, same referent -> KARMDHARAYA
 'se' meaning "by means of" = KARAN ; "from" = APADAN
 Error check order: ling, vachan, vibhakti, punarukti, word order

8.3 Exam Hall Protocol

8.3.1 The order and the clock

Your plan for the 120 minutes

0 - 10 min	HINDI (20 q)	- fastest, bank it first
10 - 22 min	ENGLISH (20 q)	- grammar and vocab are instant
22 - 30 min	GENERAL AWARENESS (20 q)	- 20 sec per question, no thinking
30 - 72 min	SECTION B, your subject (100 q)	- recall-heavy, moves fast
72 - 92 min	REASONING (20 q)	- needs working time
92 - 117 min	MATHEMATICS (20 q)	- slowest, do formula-based first
117 - 120 min	REVIEW	- check flagged questions and marking errors

If you fall behind, the questions you sacrifice are the SLOW ones, which you were least likely to get right anyway. That is the whole point of this order.

8.3.2 The rules that protect your marks

Rule	Reason
Circle NOT, FALSE, EXCEPT, INCORRECT before reading the options	A large share of these questions are negatively framed. Careless reading costs 5-10 marks per paper - more than any topic you could learn in a week
Never exceed 2 minutes on any question	One question is worth the same 1 mark as a synonym you answer in 8 seconds
Eliminate two options, then answer. Eliminate none, then skip	Expected value +0.375 versus +0.0625. This is arithmetic, not courage
In General Awareness, decide in 20 seconds	You either know it or you do not. Thinking longer does not retrieve a fact you never learned
Answer comprehension only from the passage	True-but-not-stated is the commonest trap in both language sections
For percentages, confirm the base	"20% on sales" is not "20% on cost". Wrong base, wrong answer, every time
Recheck your marking in the last 3 minutes	A misaligned answer can cost an entire section

8.3.3 The scoring arithmetic, one last time

Section	Realistic with this book	Stretch target
Reasoning	18	20
Mathematics	17	20
English	17	19
Hindi	17	19
General Awareness	14	17
Section A total	83	95

The gap between 83 and 95 is almost entirely SPEED in Maths and Reasoning, and CURRENT AFFAIRS in General Awareness. Those are the two things to work on once you know the content. Nothing else moves the number as much.

8.4 A Final Word

8.4.1 What this book can and cannot do

I have tried to give you something honest rather than something impressive.

What this book does well. Parts 4, 5, 6 and 7 - Maths, Reasoning, English and Hindi - are 80 of the 100 marks, and they are **finite and rule-based**. Every formula you need is in Chapter 8.2. Every question type is in the chapters. If you work through them with a pen and practise against a clock, 74-78 out of those 80 is a repeatable outcome, not a lucky one. That is the real engine of a high score, and it is entirely within your control.

Part 2 covers static General Awareness thoroughly, including the Delhi-specific chapter that most candidates never touch and which is the cheapest differentiator available to you.

What this book cannot do. Part 3 explains this at length and I will not soften it here: **current affairs cannot be delivered by a book**. The events had not happened when this was written. I gave you a framework, a slot table to maintain, a verification method and a daily routine - but the daily fifteen minutes has to come from you. Nobody can do that part on your behalf.

One technical limitation, restated. The Hindi Part in this PDF is in transliteration because the PDF generator I wrote cannot render Devanagari, and no Devanagari font was available in the build environment. **Use** ``hindi_section_devanagari.html`` - open it in a browser and print to PDF - for your actual Hindi practice. Chapter 7.1 explains this fully.

8.4.2 The three things that actually decide it

- 1. Section A deserves half your time, because it is half the paper.** Almost nobody gives it that, and that is precisely why it is where selections are decided. Your competition is over-preparing their subject.
- 2. Reasoning and Maths reward a pen, not eyes.** Reading a solved example teaches nothing. Cover the solution, attempt it, then compare. This single habit separates candidates who score 19 from candidates who score 12 on identical knowledge.
- 3. Start current affairs today, not next month.** Fifteen minutes daily for six months covers the window your paper will draw from. Two weeks of panic covers about 5% of it.

Work steadily. Keep the leak list honest. Revise Part 8 until the formulas come without effort.

That is enough. All the best.